MS-7788 Ver: 1.0 u-ATX : 226 * 173 mm

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CPU:

Intel - Sandy Bridge LGA 1155

System Chipset:

INTEL - Cougar Point PCH(H61)

OnBoard Chipset:

HD Audio Codec:ALC887VD / VT1708SCE
LAN:RTL 8105E 10/100 , Co-lay 8111E 10/100/1000
SIO:FIN71868AD
Flash ROM: 32Mb SPI (PCH)

Main Memory:

DDRIII (1066/1333MHz) * 2 (Dual Channel)

Expansion Slots:

PCI Express (X16) Slot * 1
PCI Express (X1) Slot * 1

PWM:

Controller: UT501 3+1 Phase
CPU+GPU: UP6282 MOSFET Driver
CPU VTT: UP1504
CPU SA : OP+MOS
DDR: UP1504
PCH: OP+MOS

ACPI:

UPI

Other:

SATA2.0 x4 (PCH)
USB2.0 RearX6 Front x4
D-SUB/DVI*1
TPM Header *1
Speaker Pin Header

0002756 3 MSI 微星內部文件 CONFIDENTIAL FOR PE(保密) ONLY (C)20111201002

MSI	MICRO-STAR INT'L CO.,LTD
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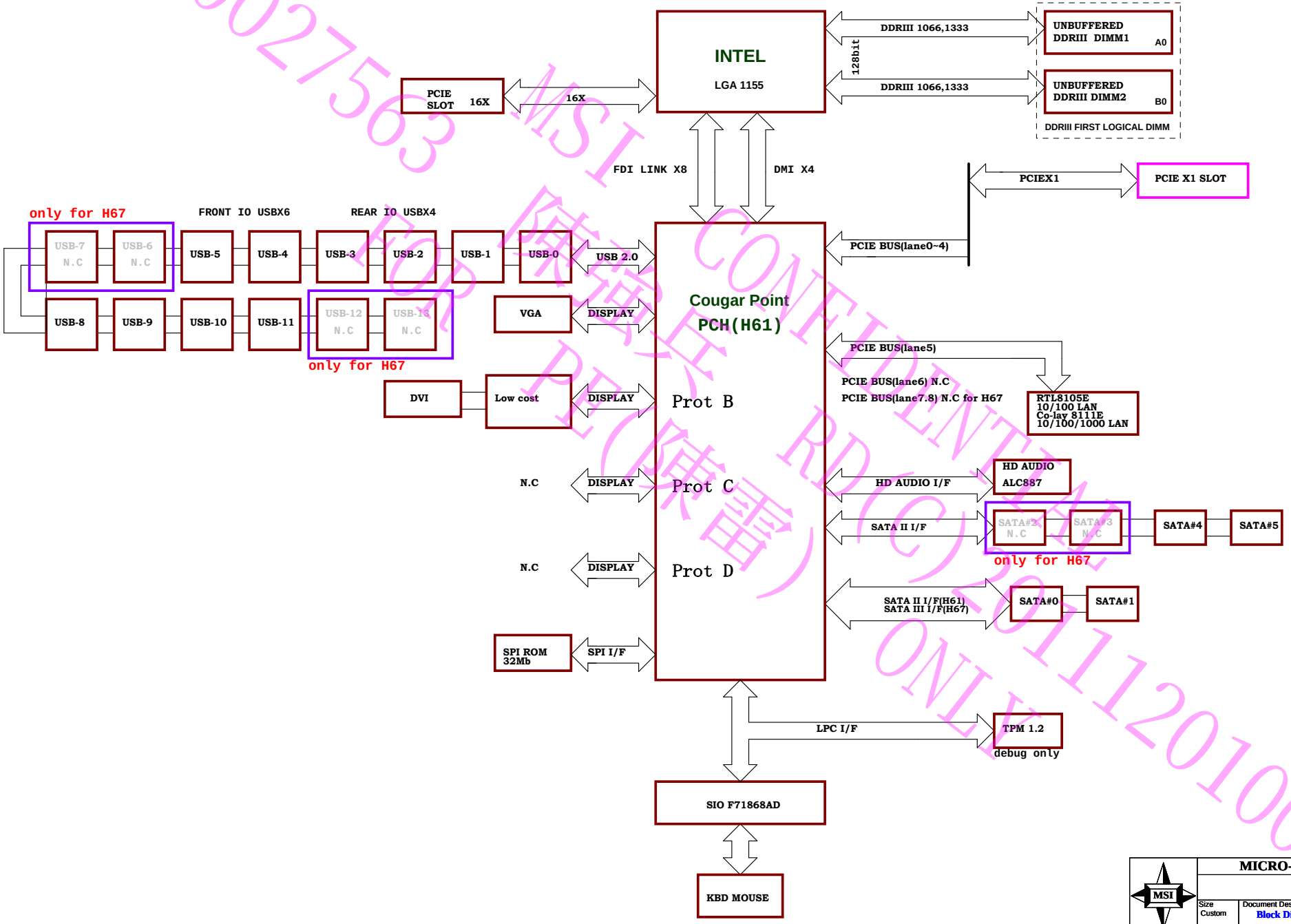
INTEL CONFIDENTIAL



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MICRO-STAR INT'L CO.,LTD	
MS-7788	
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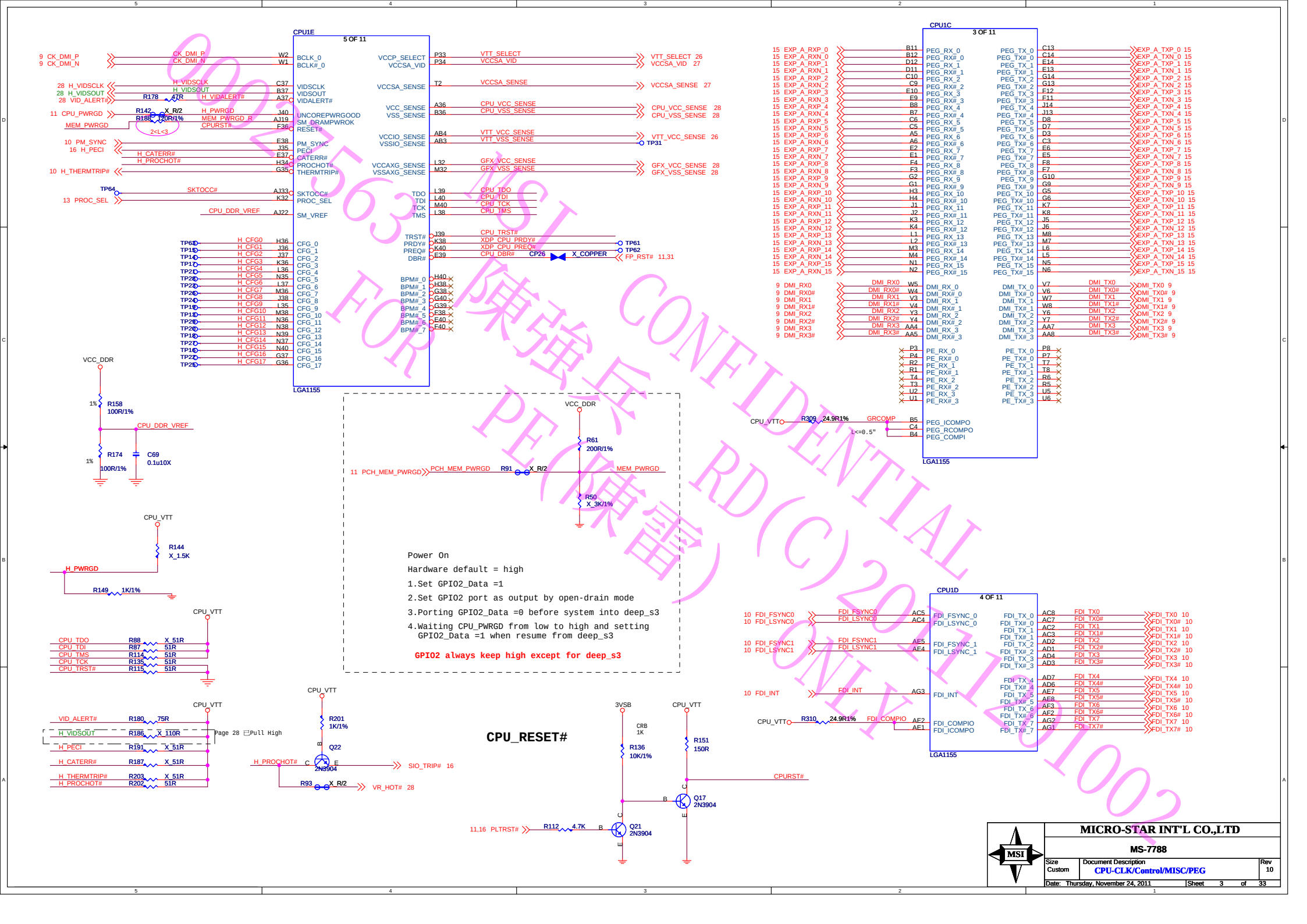
CPU1E
5 OF 11
LGA1155
CPU1C
3 OF 11
LGA1155
CPU1D
4 OF 11
LGA1155
Power On
Hardware default = high
1.Set GPIO2_Data =1
2.Set GPIO2 port as output by open-drain mode
3.Porting GPIO2_Data =0 before system into deep_s3
4.Waiting CPU_PWRGD from low to high and setting GPIO2_Data =1 when resume from deep_s3
GPIO2 always keep high except for deep_s3
CPU_RESET#
MICRO-STAR INT'L CO.,LTD
MS-7788
CPU-CLK/Control/MISC/PEG
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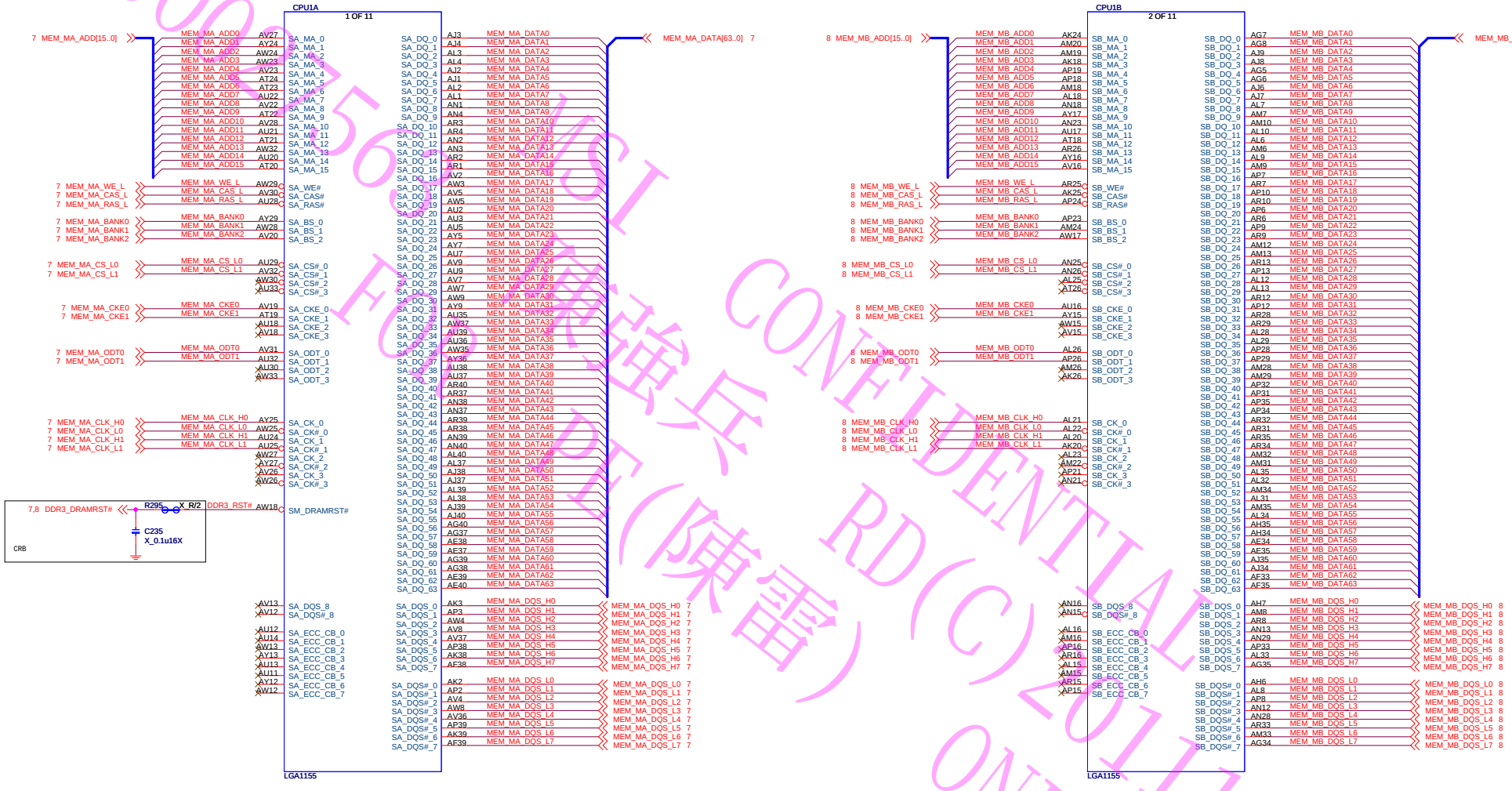

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MICRO-STAR INT'L CO.,LTD

MS-7788

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(1.05V / 1.00V)

VCCP

CPU1F
6 OF 11

A12	VCC_001	VCC_082 F32
A13	VCC_002	VCC_083 F33
A14	VCC_003	VCC_084 F34
A15	VCC_004	VCC_085 G15
A16	VCC_005	VCC_086 G16
A18	VCC_006	VCC_087 G18
A24	VCC_007	VCC_088 G19
A25	VCC_008	VCC_089 G21
A27	VCC_009	VCC_090 G22
A28	VCC_010	VCC_091 G24
B15	VCC_011	VCC_092 G25
B16	VCC_012	VCC_093 G27
B18	VCC_013	VCC_094 G28
B24	VCC_014	VCC_095 G30
B25	VCC_015	VCC_096 G31
B27	VCC_016	VCC_097 G32
B28	VCC_017	VCC_098 G33
B30	VCC_018	VCC_099 H13
B31	VCC_019	VCC_100 H14
B33	VCC_020	VCC_101 H15
B34	VCC_021	VCC_102 H16
C15	VCC_022	VCC_103 H18
C16	VCC_023	VCC_104 H19
C18	VCC_024	VCC_105 H21
C19	VCC_025	VCC_106 H22
C21	VCC_026	VCC_107 H24
C22	VCC_027	VCC_108 H25
C24	VCC_028	VCC_109 H27
C25	VCC_029	VCC_110 H28
C27	VCC_030	VCC_111 H30
C28	VCC_031	VCC_112 H31
C30	VCC_032	VCC_113 H32
C31	VCC_033	VCC_114 J12
C33	VCC_034	VCC_115 J15
C34	VCC_035	VCC_116 J16
C36	VCC_036	VCC_117 J18
D13	VCC_037	VCC_118 J19
D14	VCC_038	VCC_119 J21
D15	VCC_039	VCC_120 J22
D16	VCC_040	VCC_121 J24
D18	VCC_041	VCC_122 J25
D19	VCC_042	VCC_123 J27
D21	VCC_043	VCC_124 J28
D22	VCC_044	VCC_125 J30
D24	VCC_045	VCC_126 K15
D25	VCC_046	VCC_127 K16
D27	VCC_047	VCC_128 K18
D28	VCC_048	VCC_129 K19
D30	VCC_049	VCC_130 K21
D31	VCC_050	VCC_131 K22
D33	VCC_051	VCC_132 K24
D34	VCC_052	VCC_133 K25
D35	VCC_053	VCC_134 K27
D36	VCC_054	VCC_135 K28
E15	VCC_055	VCC_136 K30
E16	VCC_056	VCC_137 L13
E18	VCC_057	VCC_138 L14
E19	VCC_058	VCC_139 L15
E21	VCC_059	VCC_140 L16
E22	VCC_060	VCC_141 L18
E24	VCC_061	VCC_142 L19
E25	VCC_062	VCC_143 L21
E27	VCC_063	VCC_144 L22
E28	VCC_064	VCC_145 L24
E30	VCC_065	VCC_146 L25
E31	VCC_066	VCC_147 L27
E33	VCC_067	VCC_148 L28
E34	VCC_068	VCC_149 L30
E35	VCC_069	VCC_150 M14
F15	VCC_070	VCC_151 M15
F16	VCC_071	VCC_152 M16
F18	VCC_072	VCC_153 M18
F19	VCC_073	VCC_154 M19
F21	VCC_074	VCC_155 M21
F22	VCC_075	VCC_156 M22
F24	VCC_076	VCC_157 M24
F25	VCC_077	VCC_158 M25
F27	VCC_078	VCC_159 M27
F28	VCC_079	VCC_160 M28
F30	VCC_080	VCC_161 M30
F31	VCC_081	

LGA1155

CPU_VTT

CPU1H
8 OF 11

A11	VCCIO_01	VDDQ_01 AJ13
A7	VCCIO_02	VDDQ_02 AJ14
AA3	VCCIO_03	VDDQ_03 AJ20
AB8	VCCIO_04	VDDQ_04 AJ23
AF8	VCCIO_05	VDDQ_05 AJ24
AG33	VCCIO_06	VDDQ_06 AR20
AJ16	VCCIO_07	VDDQ_07 AR21
AJ17	VCCIO_08	VDDQ_08 AR22
AJ26	VCCIO_09	VDDQ_09 AR23
AJ28	VCCIO_10	VDDQ_10 AR24
AJ32	VCCIO_11	VDDQ_11 AU19
AK15	VCCIO_12	VDDQ_12 AU23
AK17	VCCIO_13	VDDQ_13 AU27
AK19	VCCIO_14	VDDQ_14 AU31
AK21	VCCIO_15	VDDQ_15 AV21
AK23	VCCIO_16	VDDQ_16 AV24
AK27	VCCIO_17	VDDQ_17 AV25
AK29	VCCIO_18	VDDQ_18 AV29
AK30	VCCIO_19	VDDQ_19 AV33
B9	VCCIO_20	VDDQ_20 AW31
D10	VCCIO_21	VDDQ_21 AY23
D6	VCCIO_22	VDDQ_22 AY26
E3	VCCIO_23	VDDQ_23 AY28
E4	VCCIO_24	
G3	VCCIO_25	
G4	VCCIO_26	
J3	VCCIO_27	
J4	VCCIO_28	VCCSA_01 H10
J7	VCCIO_29	VCCSA_02 H11
J8	VCCIO_30	VCCSA_03 H12
L3	VCCIO_31	VCCSA_04 J10
L4	VCCIO_32	VCCSA_05 K10
L7	VCCIO_33	VCCSA_06 K11
M13	VCCIO_34	VCCSA_07 L11
N3	VCCIO_35	VCCSA_08 L12
N4	VCCIO_36	VCCSA_09 M10
N7	VCCIO_37	VCCSA_10 M11
R3	VCCIO_38	VCCSA_11 M12
R4	VCCIO_39	
R7	VCCIO_40	
U3	VCCIO_41	
U4	VCCIO_42	
U7	VCCIO_43	
V8	VCCIO_44	VCCPLL_01 AK11
W3	VCCIO_45	VCCPLL_02 AK12

LGA1155

VCC_DDR

CPU_SA

VCC1_8

(0.925V / 0.85V)

+CPU_GFX

CPU1G
7 OF 11

AB33	VCCAXG_01	VCCAXG_23 T39
AB34	VCCAXG_02	VCCAXG_24 T40
AB35	VCCAXG_03	VCCAXG_25 U33
AB36	VCCAXG_04	VCCAXG_26 U34
AB37	VCCAXG_05	VCCAXG_27 U35
AB38	VCCAXG_06	VCCAXG_28 U36
AB39	VCCAXG_07	VCCAXG_29 U37
AB40	VCCAXG_08	VCCAXG_30 U38
AC33	VCCAXG_09	VCCAXG_31 U39
AC34	VCCAXG_10	VCCAXG_32 U40
AC35	VCCAXG_11	VCCAXG_33 W33
AC36	VCCAXG_12	VCCAXG_34 W34
AC37	VCCAXG_13	VCCAXG_35 W35
AC38	VCCAXG_14	VCCAXG_36 W36
AC39	VCCAXG_15	VCCAXG_37 W37
AC40	VCCAXG_16	VCCAXG_38 W38
T33	VCCAXG_17	VCCAXG_39 Y33
T34	VCCAXG_18	VCCAXG_40 Y34
T35	VCCAXG_19	VCCAXG_41 Y35
T36	VCCAXG_20	VCCAXG_42 Y36
T37	VCCAXG_21	VCCAXG_43 Y37
T38	VCCAXG_22	VCCAXG_44 Y38

LGA1155

VCCP: 112A
CPU_VTT: 8.2A
CPU_SA: 8.8A
VCC_DDR: 4.5A
VCC1_8: 1.6A

+CPU_GFX Decoupling

+CPU_GFX

C203 X_10u6.3X8
C102 22u6.3X8
C117 22u6.3X8
C109 X_22u6.3X8
C685 X_22u6.3X8
C684 X_10u6.3X8
C683 X_10u6.3X8
C681 X_10u6.3X8
C682 X_10u6.3X8

Backside

CPU SOCKET CAVITY CAPS

+1.5V_DDR3-Decoupling

VCC_DDR

C183 22u6.3X8
C201 22u6.3X8
C187 22u6.3X8
C694 X_10u6.3X8

Backside

CPU SOCKET CAVITY CAPS

CPU_SA

C708 X_10u6.3X8
C224 10u6.3X8

Backside

+CPU_VTT Decoupling

CPU_VTT

C176 10u6.3X8
C227 X_10u6.3X8
C214 X_10u6.3X8
C213 X_10u6.3X8
C226 10u6.3X8
C228 X_10u6.3X8

CPU SOCKET CAVITY CAPS

CPU_VTT

C691 X_10u6.3X8
C699 X_10u6.3X8
C693 X_10u6.3X8
C686 X_10u6.3X8
C703 X_10u6.3X8

Backside

+CPU_VCCP-Decoupling

VCCP

PLACE ALL 0805 CAPS INSIDE CPU SOCKET CAVITY

C197 10u6.3X8
C182 10u6.3X8
C215 10u6.3X8
C186 10u6.3X8
C199 10u6.3X8
C210 X_10u6.3X8
C198 X_10u6.3X8
C184 10u6.3X8
C200 10u6.3X8
C185 10u6.3X8
C211 X_10u6.3X8
C209 X_10u6.3X8
C196 X_10u6.3X8

VCC1_8

C229 10u6.3X8

C688 X_10u6.3X8
C690 X_10u6.3X8
C687 X_10u6.3X8

Backside

MICRO-STAR INT'L CO.,LTD

MS-7788

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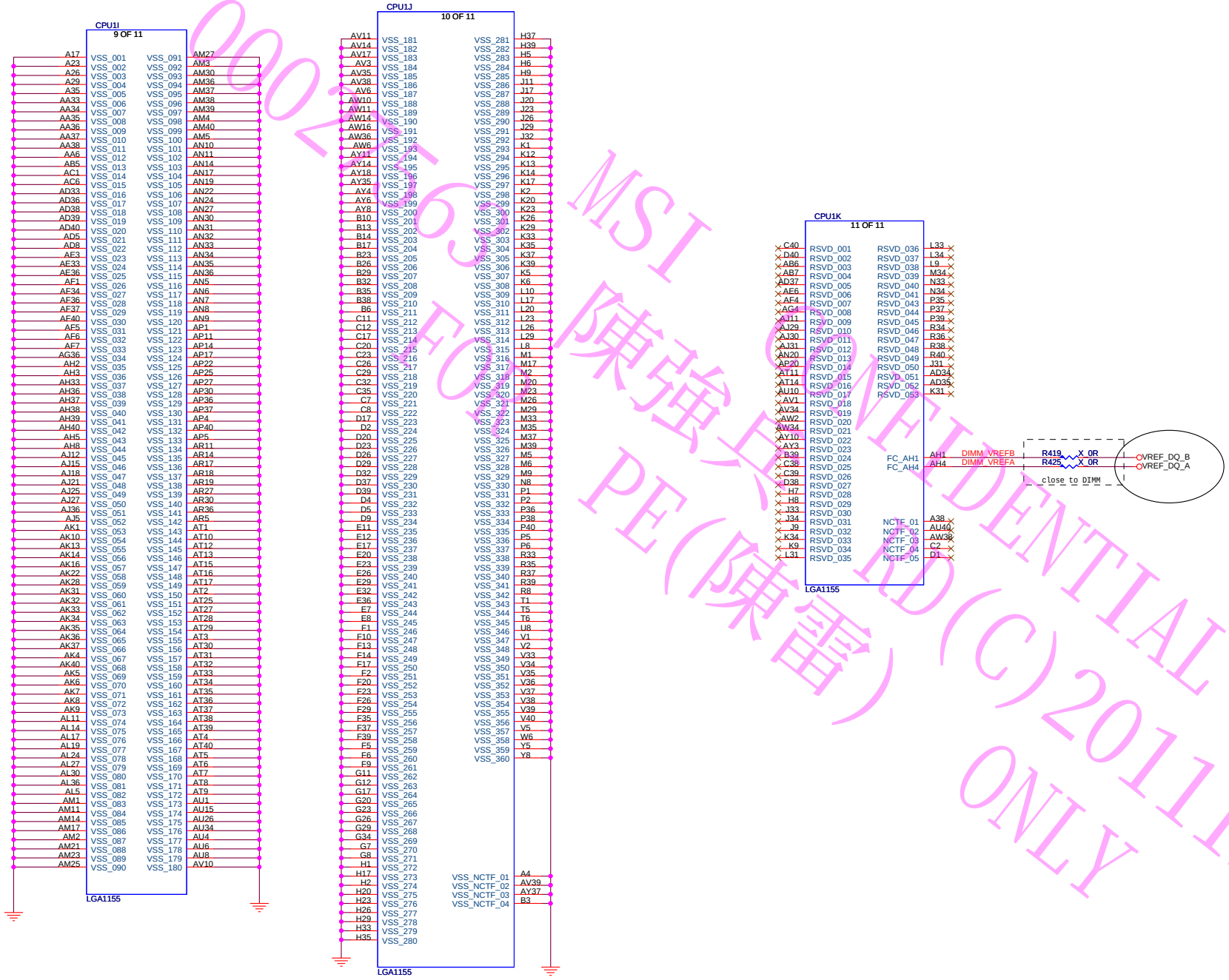

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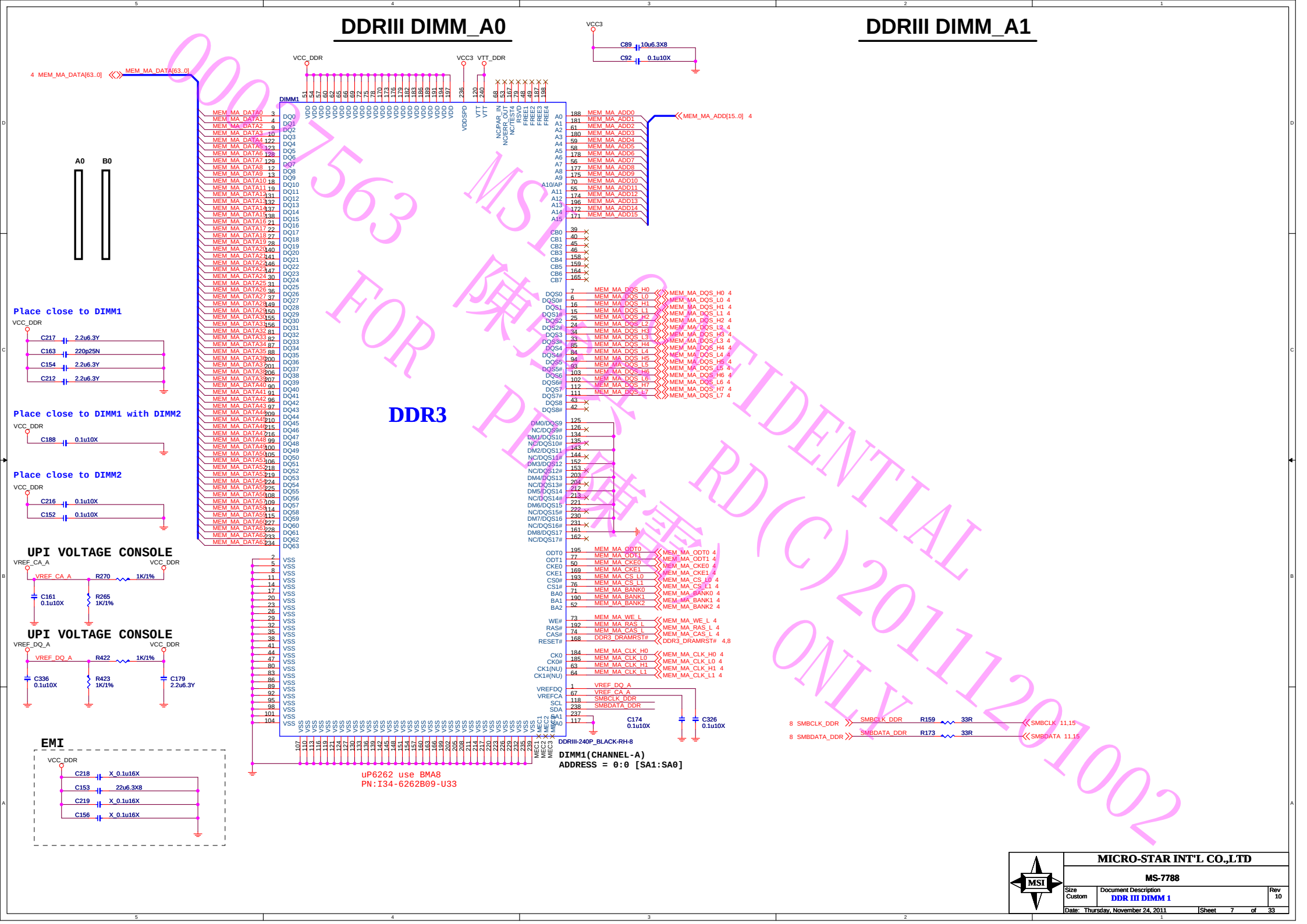
MS-7788		
Size: Custom	Document Description: CPU-GND	Rev: 10
Date: Thursday, November 24, 2011	Sheet 6 of 33	

DDRIII DIMM_A0

DDRIII DIMM_A1



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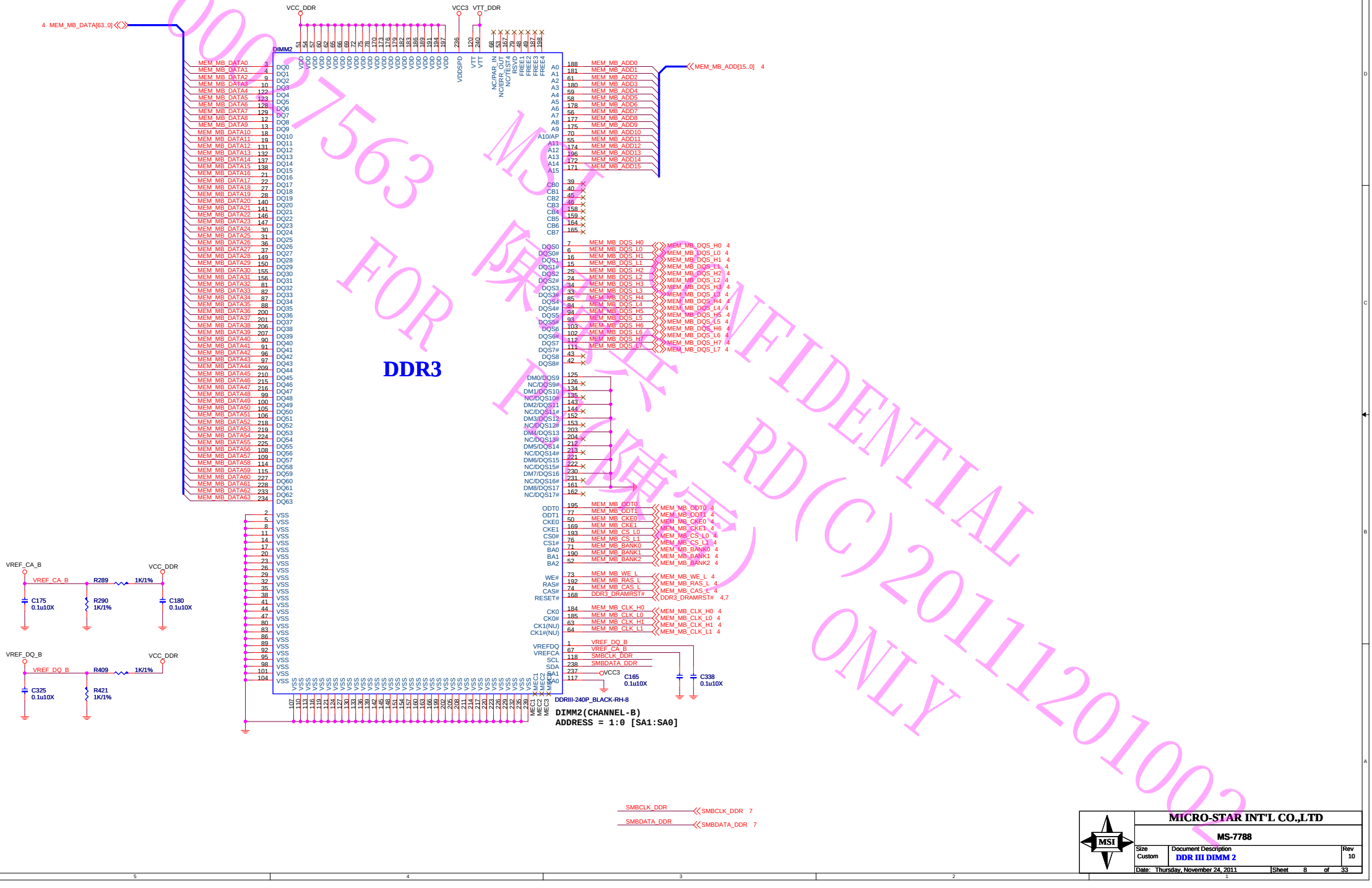
	Size: Custom	Document Description: DDR III DIMM 1	Rev: 10
	Date: Thursday, November 24, 2011	Sheet 7 of 33	

DDRIII DIMM_B0



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MICRO-STAR INT'L CO.,LTD	
MS-7788	
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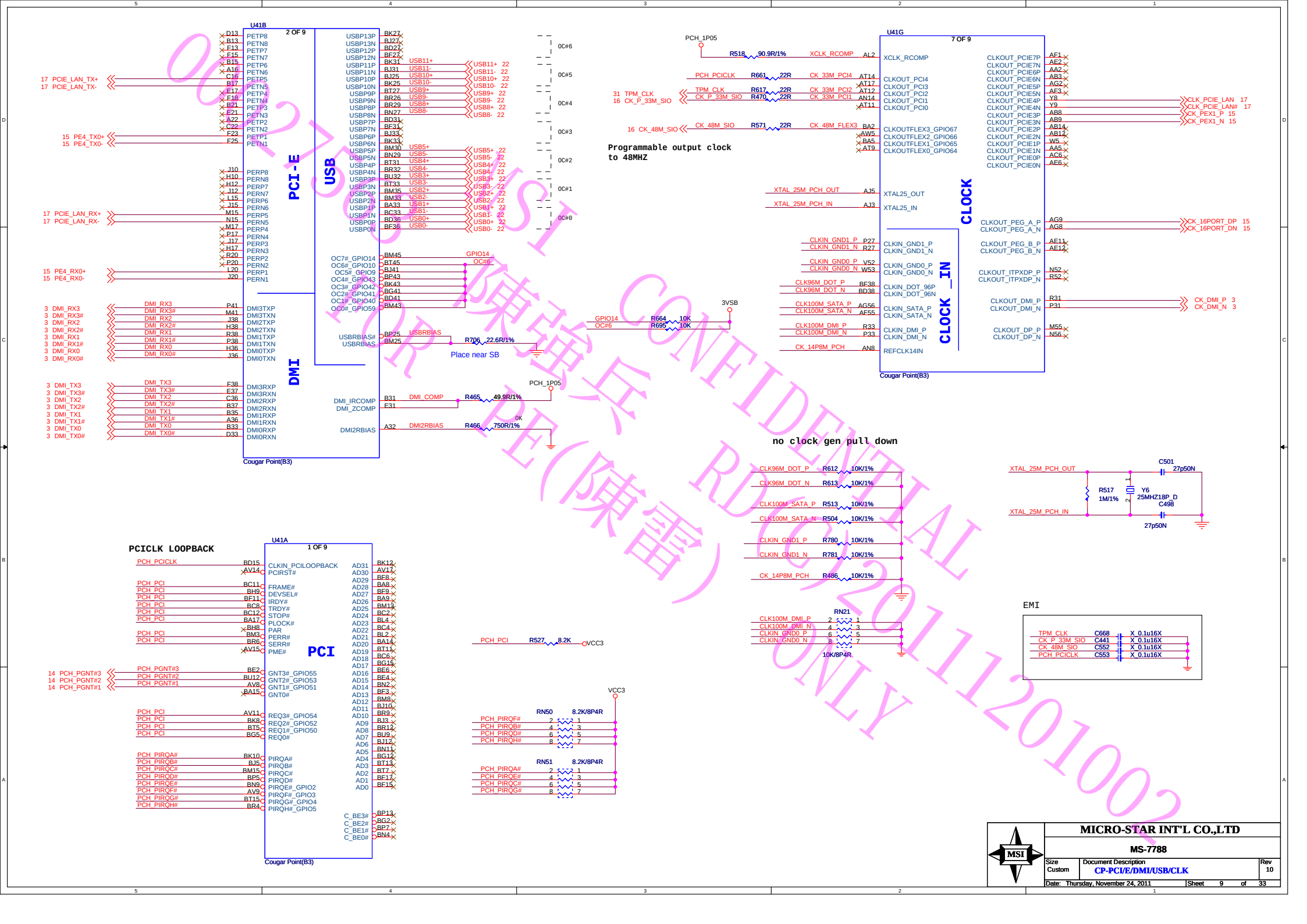
Rev: 10	
Date: Thursday, November 24, 2011	Sheet 8 of 33



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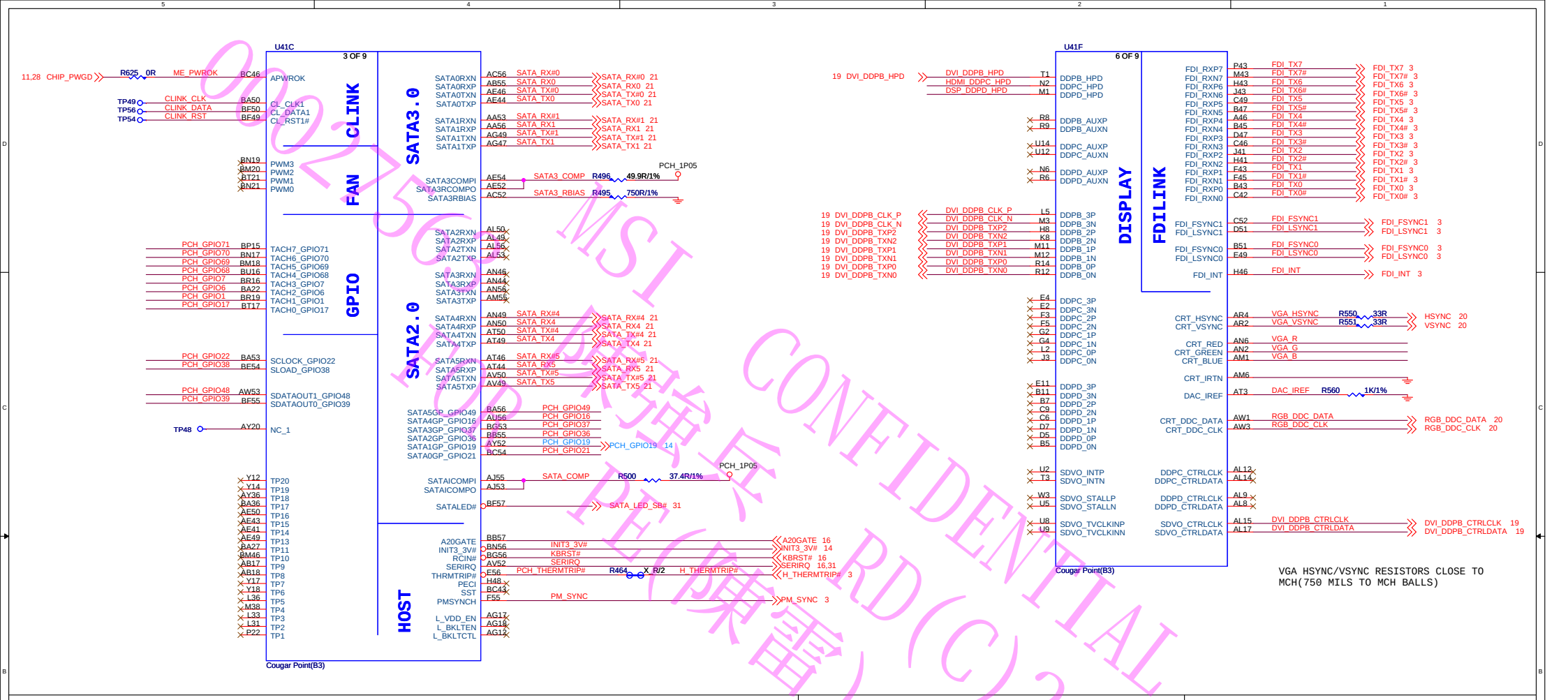
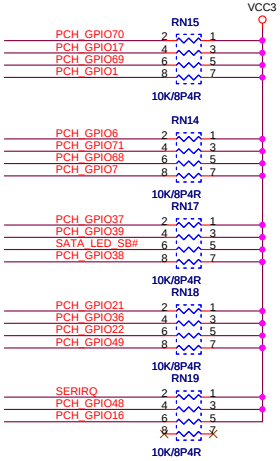
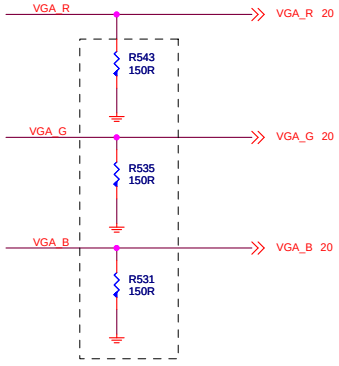


VGA HSYNC/VSYNC RESISTORS CLOSE TO MCH(750 MILS TO MCH BALLS)

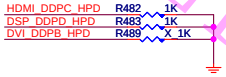
Pull HIGH for PCH



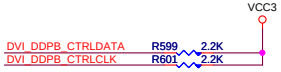


Close to PCH within 250 mils.

No VGA(pull down)



Enable VGA(CTRLCLK/DATA Pull High)



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Size: Custom	Document Description: CP-SATA/HOST/FAN/GPIO/VGA	Rev: 10
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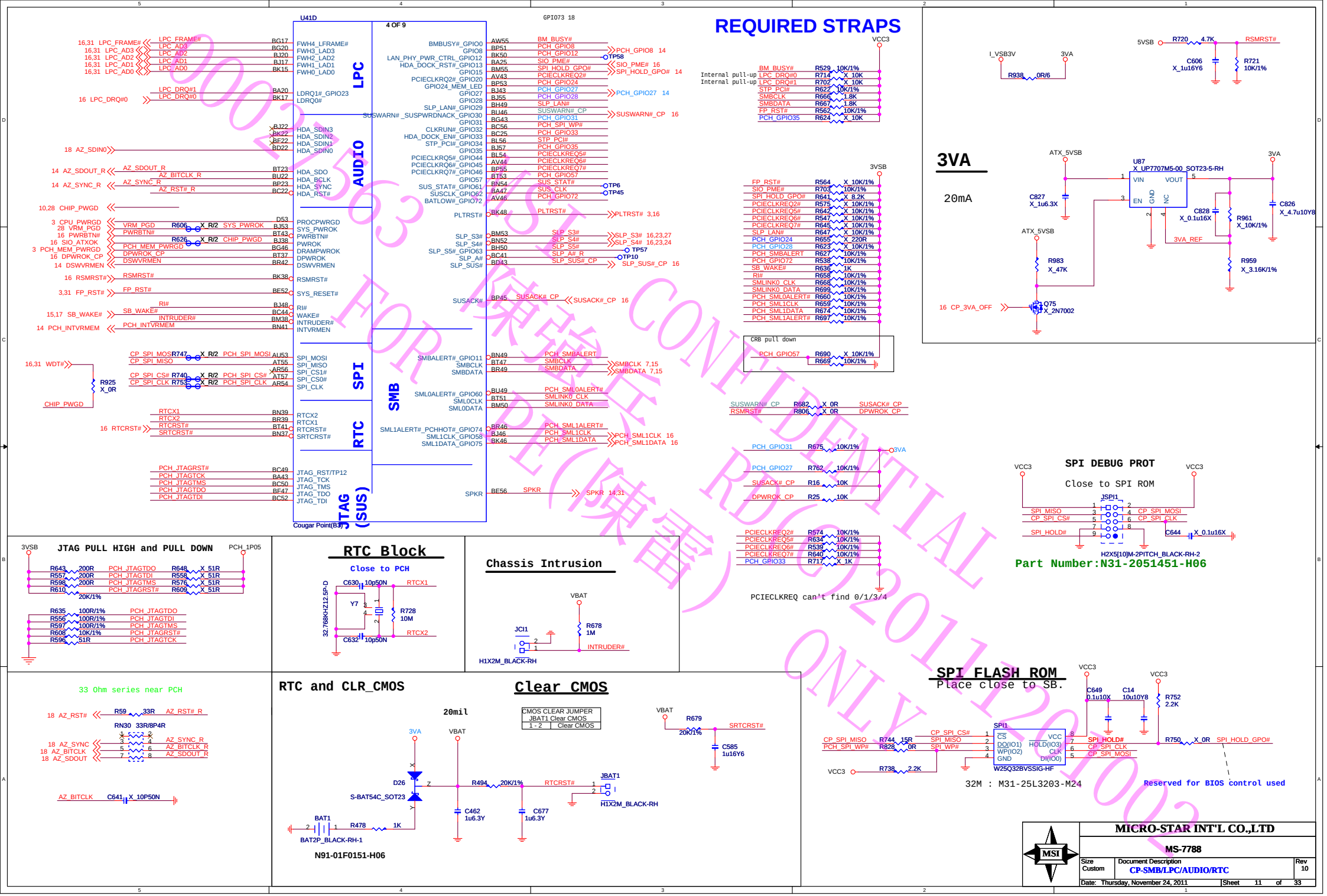
REQUIRED STRAPS
3VA
20mA
SPI DEBUG PROT
Close to SPI ROM
Part Number:N31-2051451-H06
JTAG PULL HIGH and PULL DOWN
RTC Block
Close to PCH
Chassis Intrusion
PCIECLKREQ can't find 0/1/3/4
SPI FLASH ROM
Place close to SB.
32M : M31-25L3203-M24
Reserved for BIOS control used
33 Ohm series near PCH
RTC and CLR_CMOS
Clear CMOS
20mil
N91-01F0151-H06
MICRO-STAR INT'L CO.,LTD
MS-7788
CP-SMB/LPC/AUDIO/RTC
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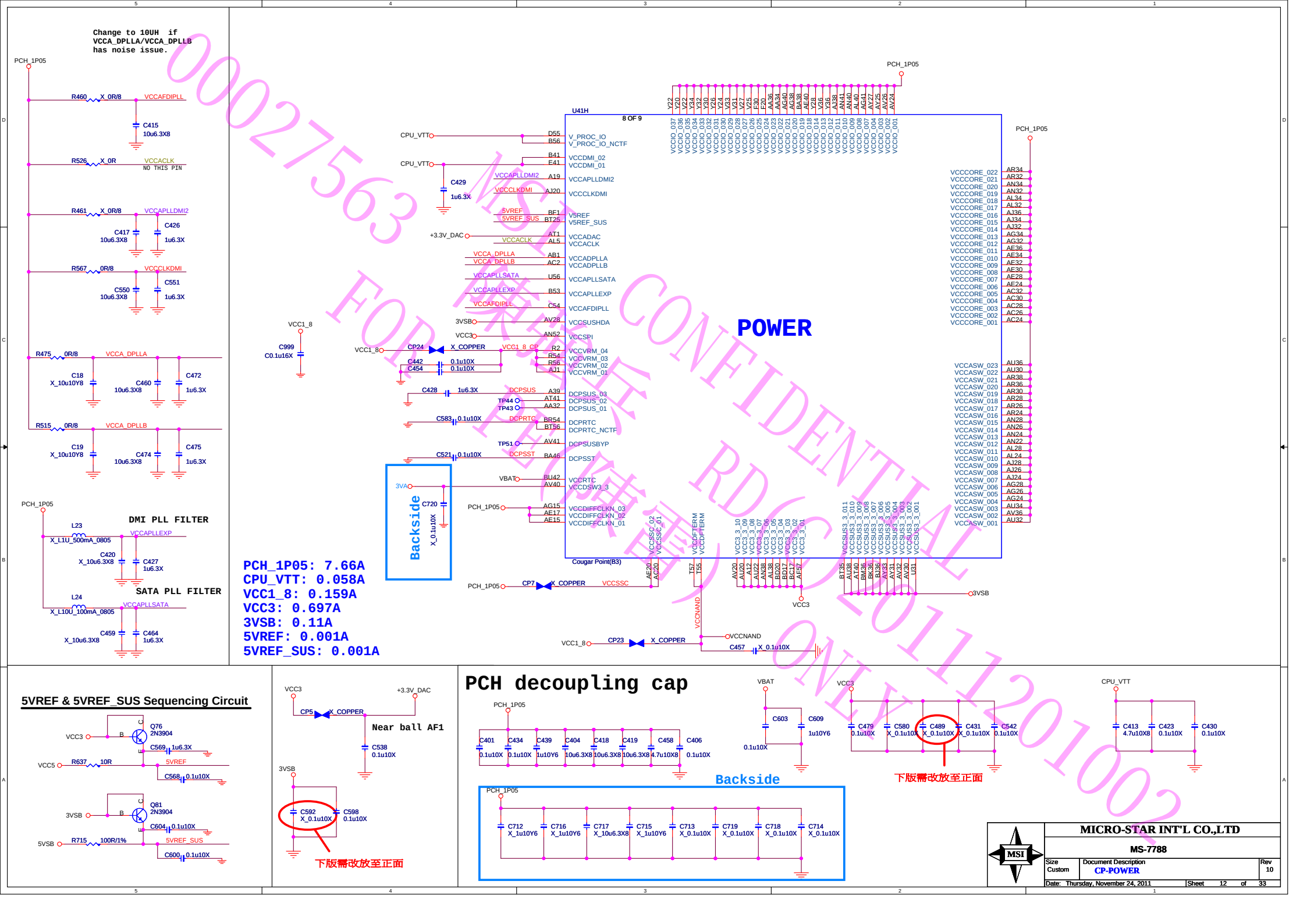
Reserved for BIOS control used


Change to 10UH if VCCA_DPLLA/VCCA_DPLLB has noise issue.
POWER
U41H
8 OF 9
Cougar Point(B3)
PCH_1P05: 7.66A
CPU_VTT: 0.058A
VCC1_8: 0.159A
VCC3: 0.697A
3VSB: 0.11A
5VREF: 0.001A
5VREF_SUS: 0.001A
DMI PLL FILTER
SATA PLL FILTER
Backside
5VREF & 5VREF_SUS Sequencing Circuit
Near ball AF1
PCH decoupling cap
Backside
下版需改放至正面
下版需改放至正面
MICRO-STAR INT'L CO.,LTD
MS-7788
CP-POWER
Rev 10
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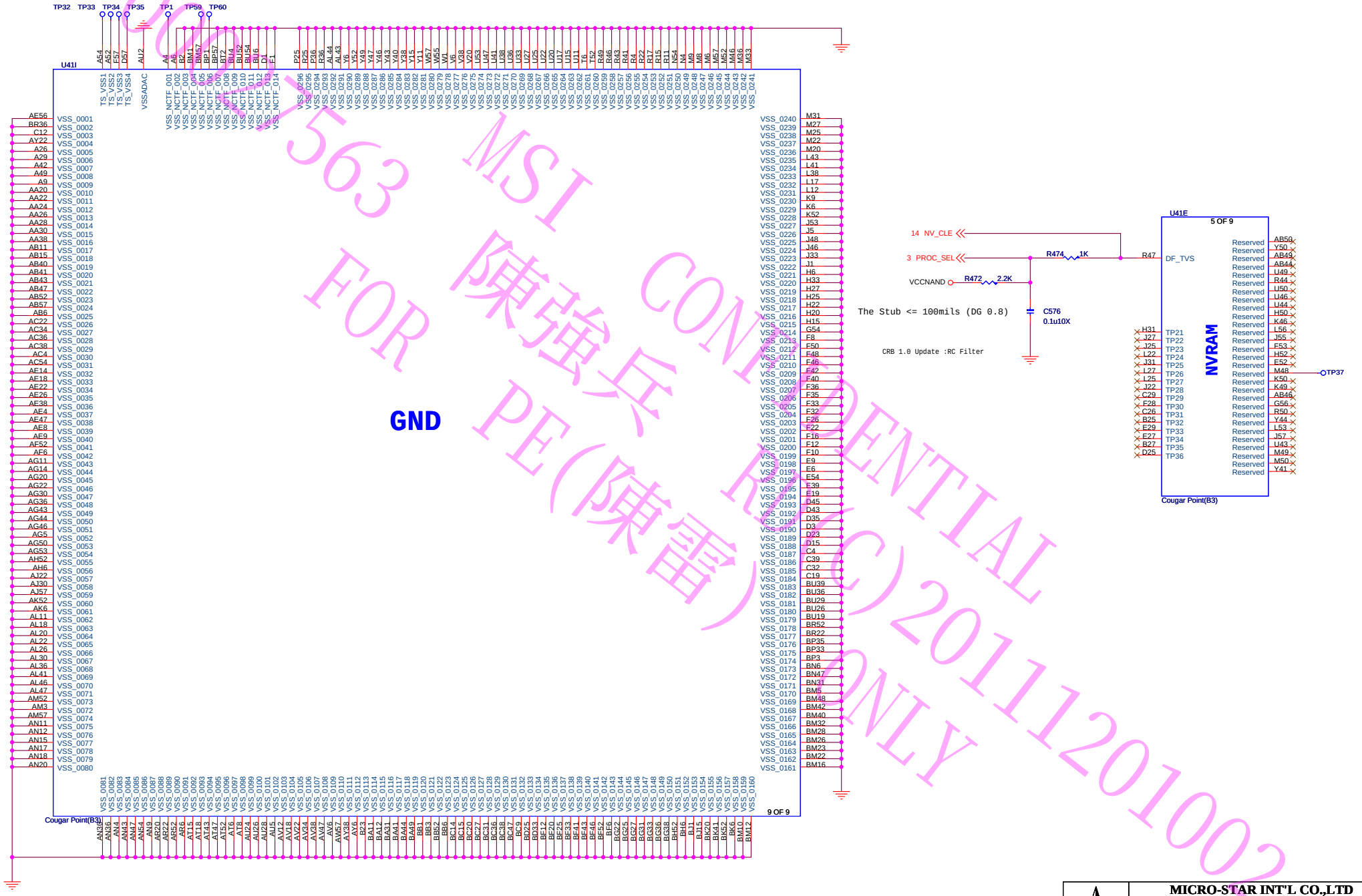

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MS-7788		
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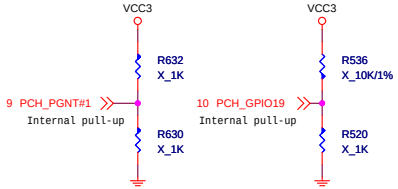
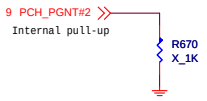
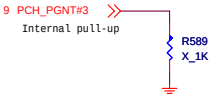
PCH Straps

BOOT DEVICE	GNT1	SATA1GP/GPIO19
LPC	0	0
PCI	1	0
SPI	1	1

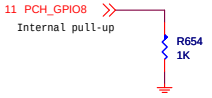




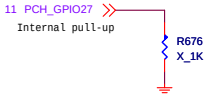
DMI AC/DC MODE
0 : AC
1 : DC *



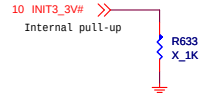
Topblock swap override when pull-low
Signal has a weak internal pull-up



GPIO8
0 : Integrated Clocking Enable (FCIM)*
1 : Buffer Through Mode Enable (BTM)

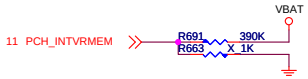


GPIO28
0 : OD PLL VR disabled
1 : OD PLL VR enabled *
Signal has a weak internal pull-up



INT3_3V#
0 : ??????????????
1 : ?????????????? *

1: INIT3_3V to asserted for 16 PCI clock to reset the processor by some evens occur.
0: Can not to reset the processor.

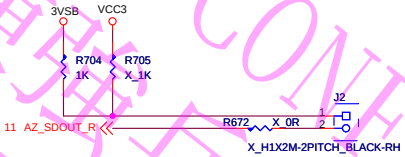


INTVRMEN
0: DISABLE INTERNAL VRM
1: ENABLE INTERNAL VRM *

When these voltageregulators are enabled, the integrated GbE only operates at 10/100 Mbps during S3-S5.

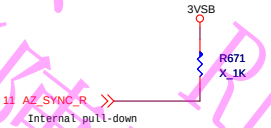
DSWVRMEN
0 : Disable Internal Deep Sleep 1.05 V regulators.
1 : Enable Internal Deep Sleep 1.05 V regulators.

This signal enables the internal Deep Sleep 1.05 V regulators. Must beconnected even when not supporting DSW.



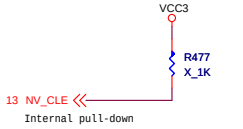
HDA_SDO
Disable ME in Manufacturing Mode
when pull LOW ????

HDA_SDO has internal pull down.
Default should be connected to SDIN of codec, no pull up/down.
To Disable ME need to have a jumper to pull high

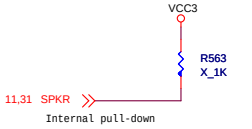


HDA_SYNC
OD PLL VR SUPPLY SEL
0: 1.8V SUPPLY *
1: 1.5V SUPPLY

GPIO15
0 : TLS CIPHER SUITE WITH NO CONFIDENTIALITY *
1 : TLS CIPHER SUITE WITH CONFIDENTIALITY



DMI/FDI TERMINATION VOLTAGE
DC COUPLED: TX/RX TO VCC ISF SAMPLED HIGH
DC COUPLED: TX/RX TO VSS IF SAMPLED LOW *?
AC COUPLED: TX SET TO VCC/2, RX SET TO VSS REGARDLESS OF THIS STRAP



SPKR
0 : EN TCO REBOOT *
1 : DIS TCO REBOOT

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PCI_Express X16 Slot

N11-1640971-K06

Trace width > 200 mils

PCI_E1

SLOT-PCI164P_BLUE-2PITCH-RH

PCI EXPRESS x1-PORT

PCI_E2

SLOT-PCI36P_BLACK-2PITCH-RH-10

0002156 PCI 廠商保密 CONFIDENTIAL RD(C)20111201002 ONLY

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MS-7788		
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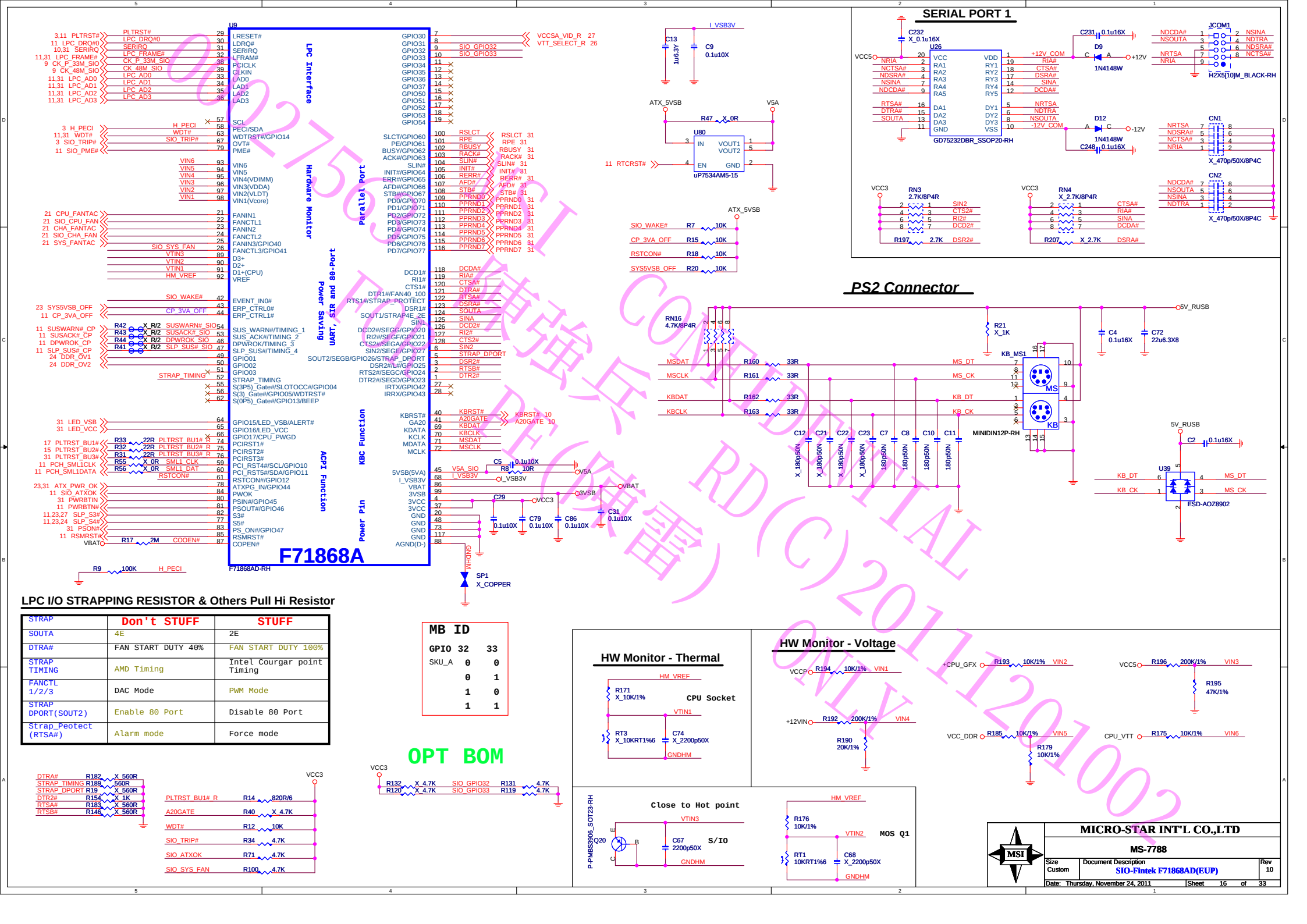

LPC I/O STRAPPING RESISTOR & Others Pull Hi Resistor

STRAP	Don't STUFF	STUFF
SOUTA	4E	2E
DTRA#	FAN START DUTY 40%	FAN START DUTY 100%
STRAP TIMING	AMD Timing	Intel Courgar point Timing
FANCTL 1/2/3	DAC Mode	PWM Mode
STRAP DPORT(SOUT2)	Enable 80 Port	Disable 80 Port
Strap_Peotect (RTSA#)	Alarm mode	Force mode

MB ID

GPIO	32	33
SKU_A	0	0
	0	1
	1	0
	1	1

OPT BOM

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RTL8105E 10/100M LAN
co-lay RTL8111E Giga LAN

Power Requlator Option

	Mode	Choke6	C394	C397	C414	C421	R262	R263
8111E Series & 8105E Series	SWR mode	O	O	O	O	O	O	X
8105E Series	LDO mode	X	X	X	X	X	X	O

VDD33 — R262 X_0R — R263 0R — ENSWREG

width>60mil CHOKE6 X_4.7u0.75A190m

VDD10 — VDD10 — 1 — 2 — REGOUT

C394 X_10u6.3X8 — C397 X_0.1u10X

near pin36 <200mil

CHOKE(>0.6A) AVL:
L04-47A7340-T04

VCC3 — R476 1K — R471 15K1%/4 — LAN_ISO

11,15 SB_WAKE#

R456 2.49K/1% RSET — ENSWREG

width>40mil

VDD33 — C414 X_4.7u6.3X8 — C421 X_0.1u10X

REGOUT

near pin <200mil

U38

9 PCIE_LAN_TX+ — C443 0.1u10X — PCIE_LAN_TXP_C 17 — HSIP
9 PCIE_LAN_TX- — C444 0.1u10X — PCIE_LAN_TXN_C 18 — HSIN

9 CLK_PCIE_LAN — CLK_PCIE_LAN 19 — REFCLK_P
9 CLK_PCIE_LAN# — CLK_PCIE_LAN# 20 — REFCLK_N

PCIE interface

22 PCIE_LAN_RXP_C — C447 0.1u10X — PCIE_LAN_RX+ 9 — HSOP
23 PCIE_LAN_RXN_C — C448 0.1u10X — PCIE_LAN_RX- 9 — HSON

25 PERSTB — PLTRST_BU1# 16
16 CLKREQB

LAN_ISO 26 — ISOLATEB
28 — LANWACKEB

PM

RSET 46 — RSET
ENSWREG 33 — ENSWREG

Regulator

VDD33 — 34 — VDDREG
35 — VDDREG
36 — REGOUT

27 — DVDD33
39 — DVDD33

POWER

VDD33 — 42 — AVDD33
47 — AVDD33
48 — AVDD33
12 — AVDD33(NC)

VDD10 — 13 — DVDD10
29 — DVDD10
41 — DVDD10(NC)
3 — AVDD10

R480 0R/6

45 — AVDD10
6 — AVDD10(NC)
9 — AVCC10(NC)

EVDD10 — 21 — EVDD10

C451 1u6.3X — C445 0.1u10X

GND — 24
GND Pad — 49

RTL8105E-VL-CG

銅製程

Transceiver Interface

1 — MDIP0 — TR_D0+
2 — MDIN0 — TR_D0-
4 — MDIP1 — TR_D1+
5 — MDIN1 — TR_D1-
7 — MDIP2(NC) — TR_D2+
8 — MDIN2(NC) — TR_D2-
10 — MDIP3(NC) — TR_D3+
11 — MDIN3(NC) — TR_D3-

EEPROM

40 — LED0 — LED0_LINK100#
37 — LED1/EESK — LAN_EESK
30 — EECS/SCL — LAN_EECS — R467 10K
32 — EEDI/SDA — LAN_EEDI — R462 10K
31 — LED3/EEDO — LAN_EEDO

GPO:
1: Link up
0: Link down

38 GPO — R457 1K — VDD33

Keep R457 for RTL8105E-VB application
Remove R457 for RTL8105E Series (except for VB)

14 — SMBCLK(NC)
15 — SMBDATA(NC) — LAN_SMB_DA — R473 X_10K

8111E: stuff
8105E: unstuff

CLOCK

43 CKXTAL1 — CLK_LANI — C395 27p50N
Y2 25MHZ18P_D
44 CKXTAL2 — CLK_LANO — C398 27p50N

Pin49: 9 via from top layer to GND layer
and make the via at the center of IC.

For RTL8111E Series EEPROM / ASF / Efuse Function

	EEPROM 93C46	EEPROM 93C56/66	EEPROM 93C46	EFuse
SMBCLK	X	X	L	X
SMBDATA	L	H	H	L
GPO	H	H	H	H

Note: For RTL8111E-VL, R17 must be 1K.

3.3v Power on rise time : 1~100ms. MAX: 163mA

Place near pin

VDD33 — 3VSB — CP14 X_COPPER — CP6 X_COPPER

27 — C436 0.1u10X
39 — C409 0.1u10X
42 — C407 0.1u10X
47 — C411 0.1u10X
48 — C410 0.1u10X
12 — C438 X_0.1u10X

8111E: stuff
8105E: unstuff

Place near pin

VDD10

3 — C416 0.1u10X
13 — C446 0.1u10X
19 — C433 0.1u10X
45 — C412 0.1u10X
41 — C408 X_0.1u10X
6 — C424 X_0.1u10X
9 — C435 X_0.1u10X

8111E: stuff
8105E: unstuff

LAN Connector

VDD33 — EMI — C162 X_0.1u10X

LAN_USB1B

19		
20		
13	PWR	
18	TD1+	TR_D0+
12	TD1-	TR_D0-
17	TD2+	TR_D1+
11	TD2-	TR_D1-
16	TD3+	TR_D2+
10	TD3-	TR_D2-
15	TD4+	TR_D3+
9	TD4-	TR_D3-
14	GND	GND/RCT
21		LED1_LINK1000#
22		LED0_LINK100#

RJ45_USBX2_LEDX2_TX-100-RH-7

8111E: unstuff
8105E: stuff

TCT — C202 0.01u10X

GND/RCT — C158 0.01u10X — R261 X_0R

8111E: 0R
8105E: 0.01uF

LAN_EEDO — R267 510R

8111E: 200R
8105E: 510R

LED3_ACT
TCT
TR_D0+
TR_D0-
TR_D1+
TR_D1-
TR_D2+
TR_D2-
TR_D3+
TR_D3-
GND/RCT

8111E: unstuff
8105E: 510R

VDD33 — R271 510R

LAN_EESK — R277 X_200R

LED1_LINK1000#
LED0_LINK100#

8111E: 200R
8105E: unstuff

only support LED0+LED1/LED1+LED3 dual color LED combinations when using EEPROM

Giga-Lan		10/100-Lan	
N58-22F0731		N58-22F0771	
Link	Yellow	Link	Yellow
Active	Blinking	Active	Blinking
1000	Orange	100	Green
100	Green	10	None
10	None		

19 — 250R — Yellow
20

21 — Orange
22 — 250R — Green

19 — Yellow
20

21 — Green
22

LED3_ACT
LED1_LINK1000#
LED0_LINK100#

C157 0.1u10X — C168 0.1u10X — C181 0.1u10X

8105E POWER Consumption

	3.3V	mW
10 M Idle/TxRx	14/75	46/248
100 M Idle/TxRx	43/66	142/218
S0 ALDPS	3.2	11

8111E POWER Consumption

	3.3V	mW
10 M Idle/TxRx	12/66	40/218
100 M Idle/TxRx	31/44	102/145
Giga Idle/TxRx	135/163	452/538
ALDPS	4	13

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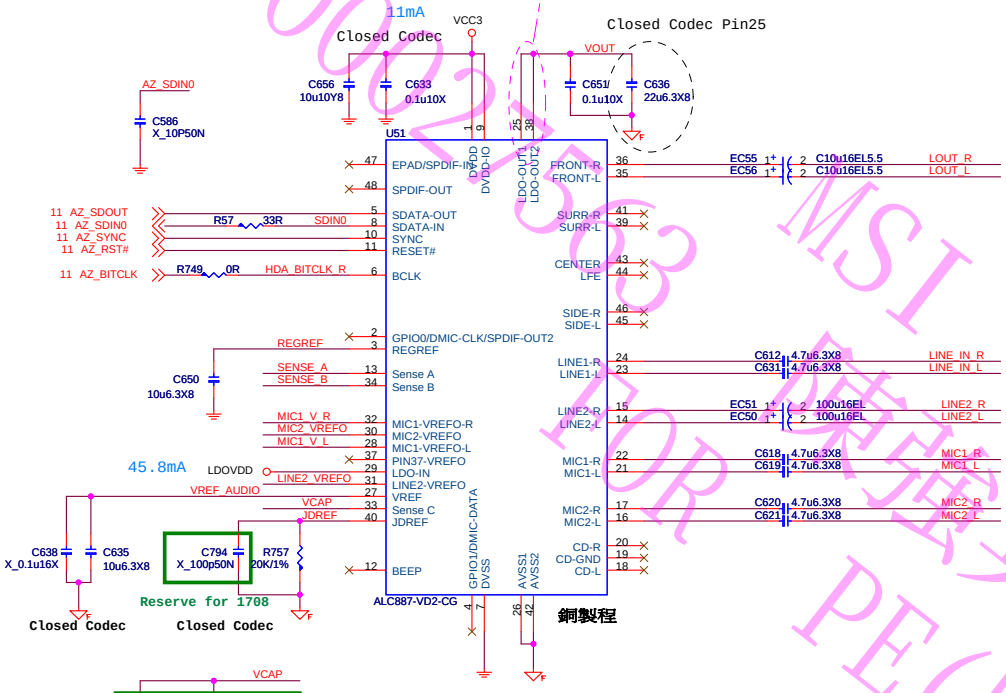
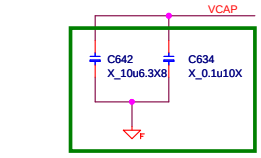
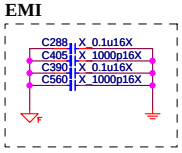
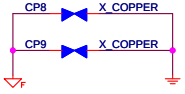
ALC887-VD
VT1708S CE

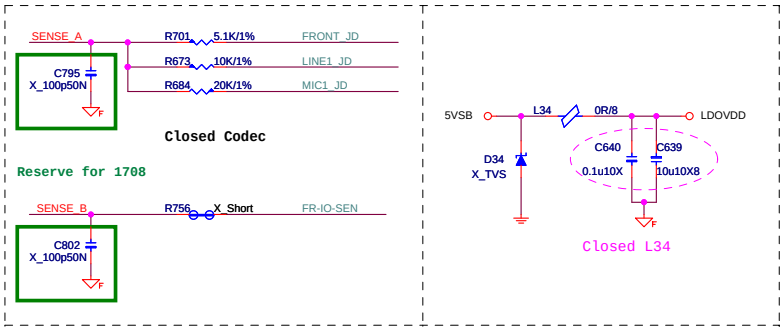
Codec Pin25 & 38連接的Layout，以最短路徑，至少40mils線寬連接。



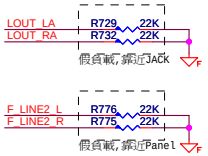


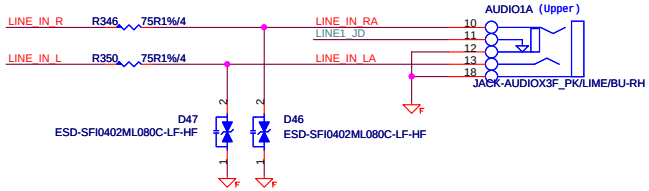


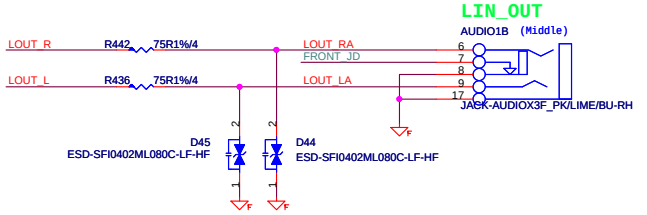


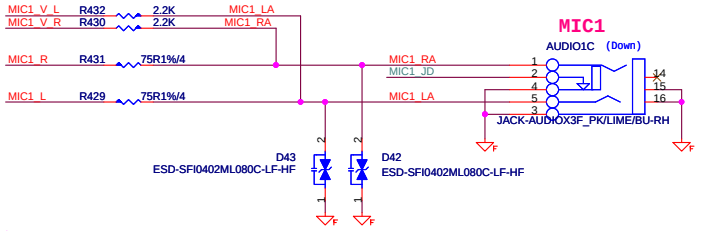


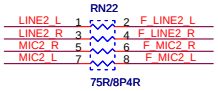
當串接電容有極性時，需上對地電阻

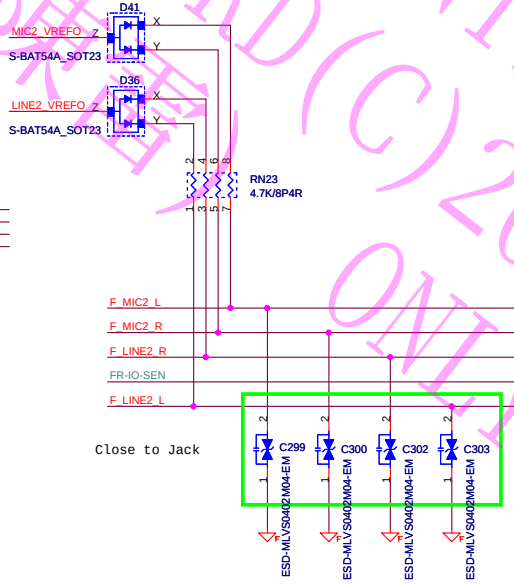


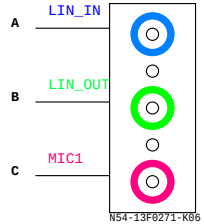


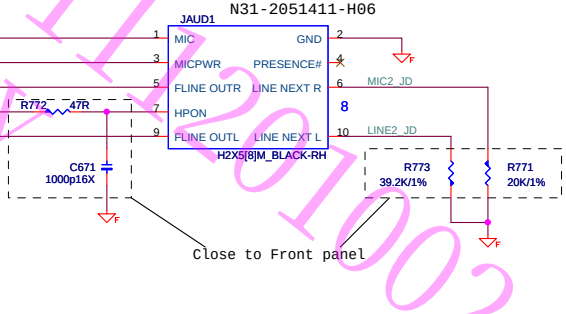












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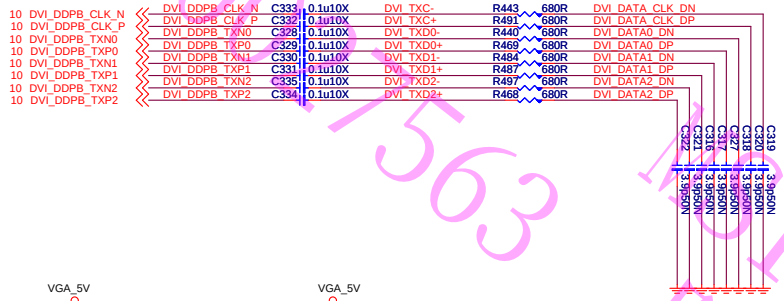
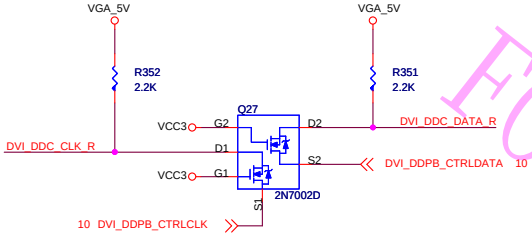
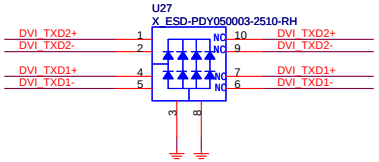
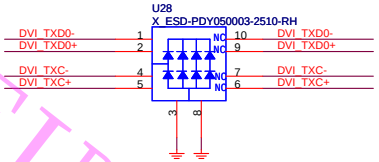
DVI level shifter

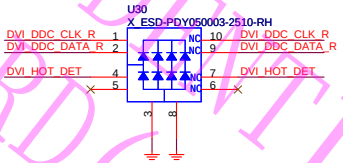
VGA: resolution of 2048x1536 pixels with 32-bit color at 75 Hz (4:3 QXGA)

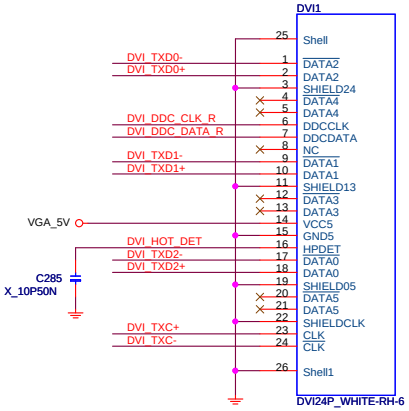


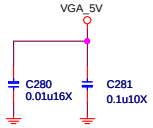


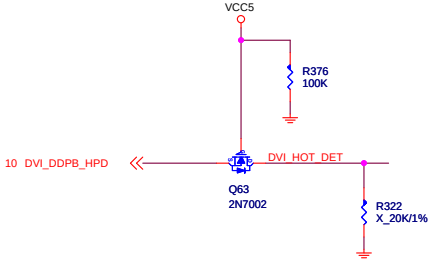


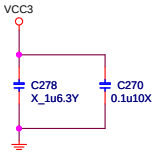


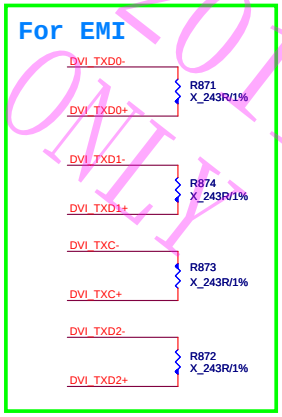


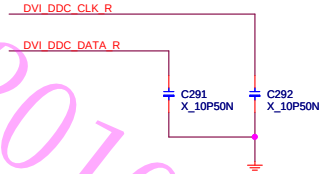












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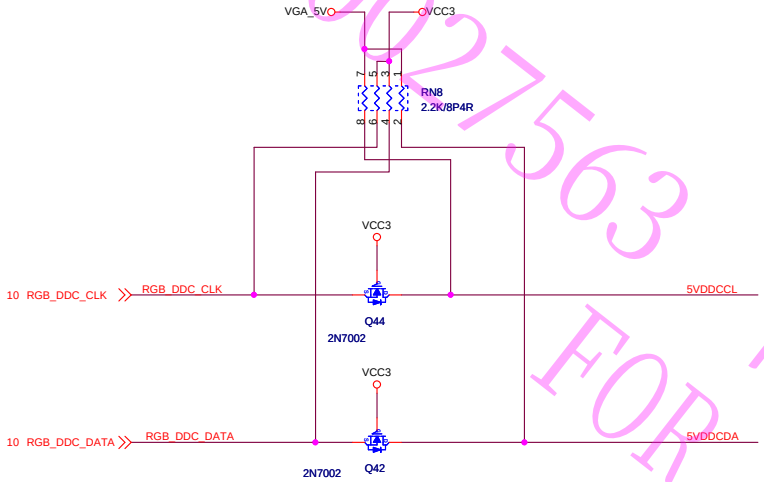
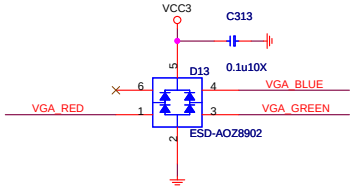
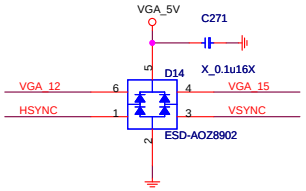
D-Sub

VGA: resolution of 2048x1536 pixels with 32-bit color at 75 Hz (4:3 QXGA)

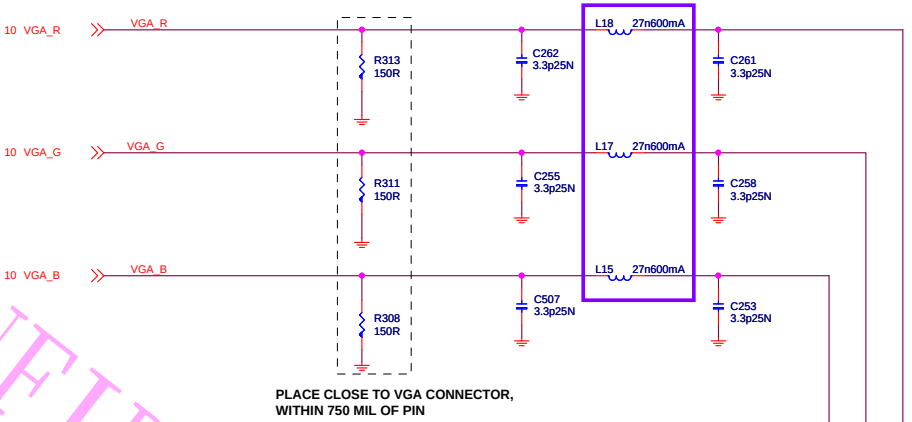
Level shift







Change to L, don't use FB
Try to pass VESA v1.2



PLACE CLOSE TO VGA CONNECTOR,
WITHIN 750 MIL OF PIN

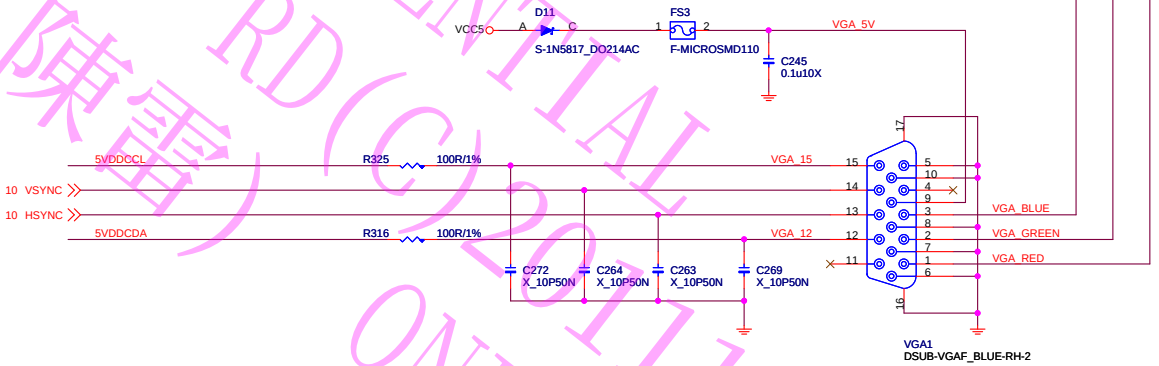


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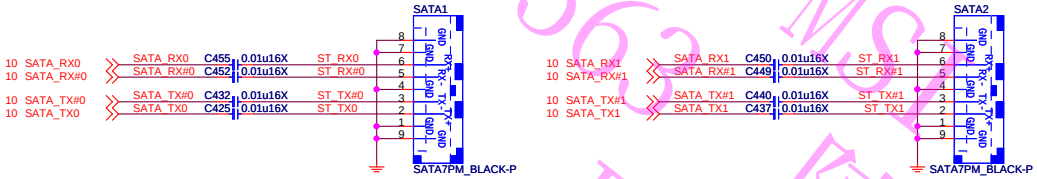
Table 1-3. Desktop Intel® 6 Series Chipset SKUs

Feature Set	SKU Name(s)					
	Q67	Q65	B65	H67	P67	H61
Total number of SATA ports	6	6	6	6	6	4
• SATA Ports (6 Gb/s, 3 Gb/s, and 1.5 Gb/s)	2^4	1^5	1^5	2^4	2^4	0
• SATA Ports (3 Gb/s and 1.5 Gb/s only)	4	5	5	4	4	4^8

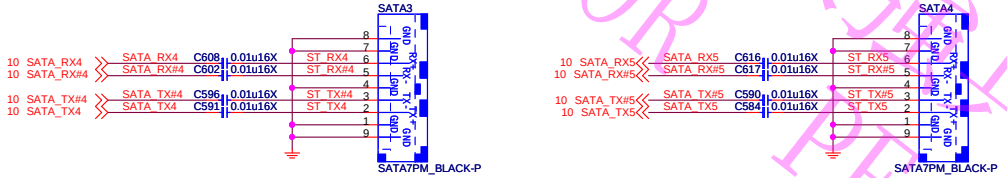
NOTES:

8. SATA ports 2 and 3 are disabled.

SATA 3G PORT 0,1

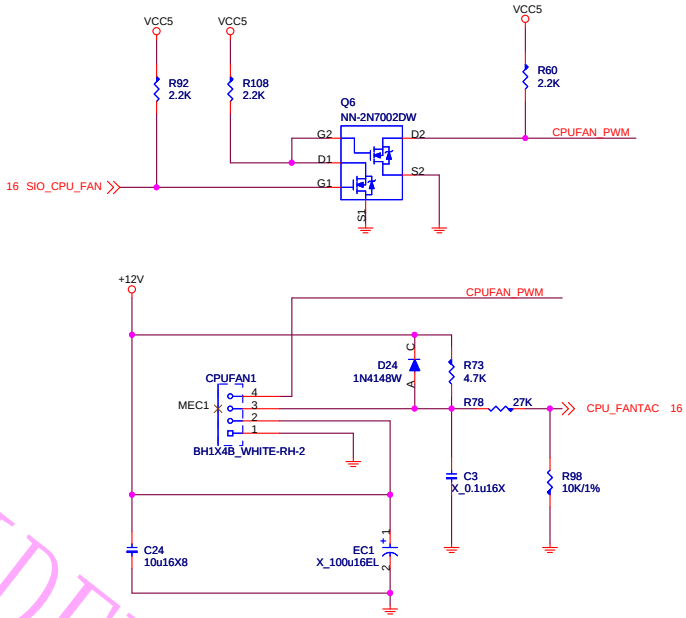


SATA 3G PORT 4,5



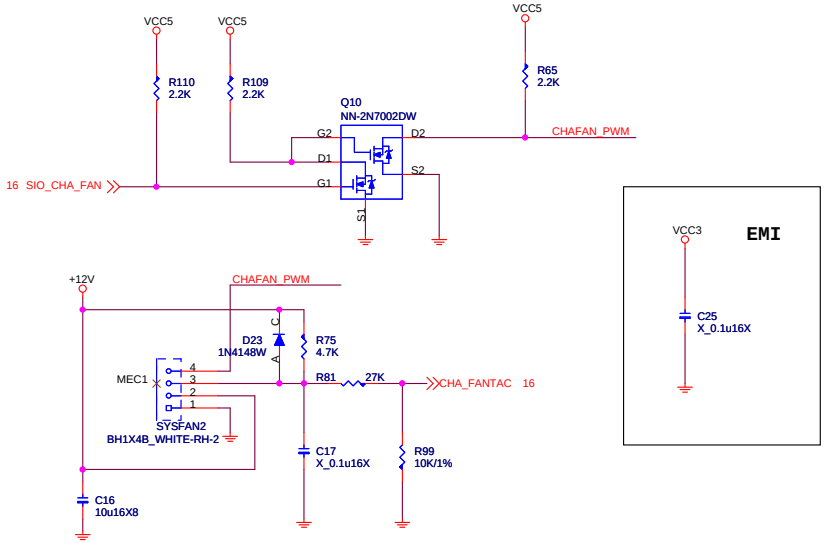
CPU FAN-COUNTROL CIRCUIT

From SIO SYSTEM FAN1



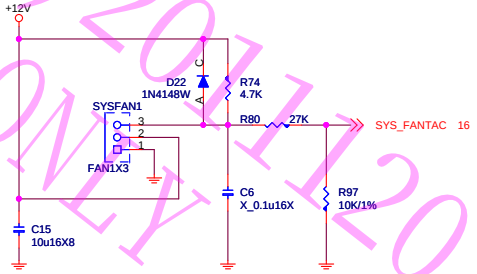
CHASSIS FAN-COUNTROL CIRCUIT

From SIO SYSTEM FAN2



SYSTEM FAN3-COUNTROL CIRCUIT

From SIO SYSTEM FAN3



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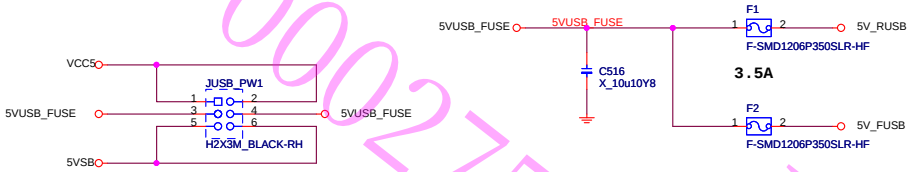
5V_RUSB Switch

5V_RUSB must 120mm



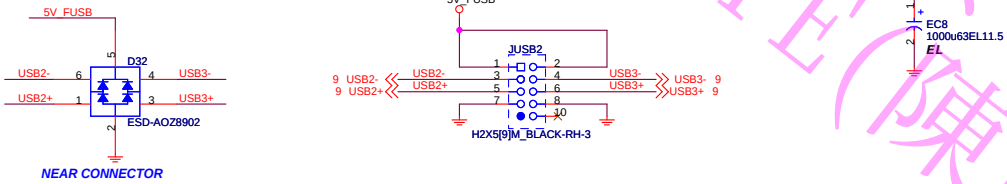
Default VCC5 (PIN1-3,2-4)

JUSB_PW1	BIOS Menu	Wake up support
1-3,2-4	EUP Enable	Not support
	EUP Disable	
3-5,4-6	EUP Enable	Not support
	EUP Disable	support

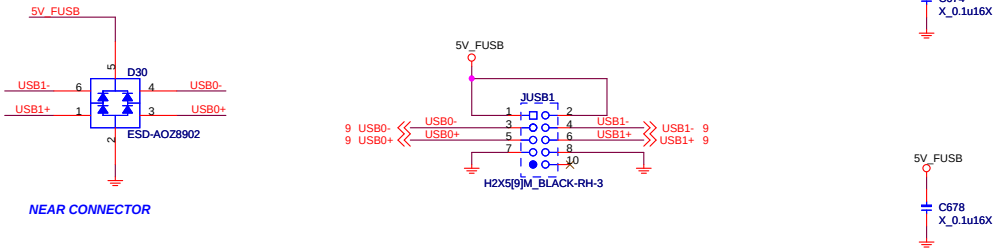
Front USB Connector

For H61 6,7,12,13 Port should be remove

FRONT USB PORT 2,3

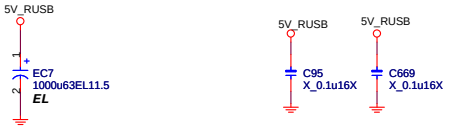


FRONT USB PORT 0,1

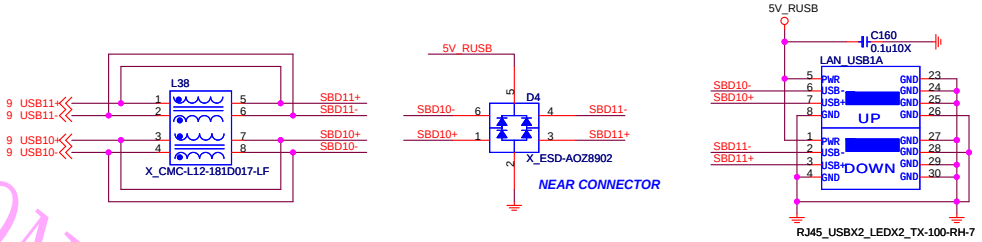


Rear USB Connector

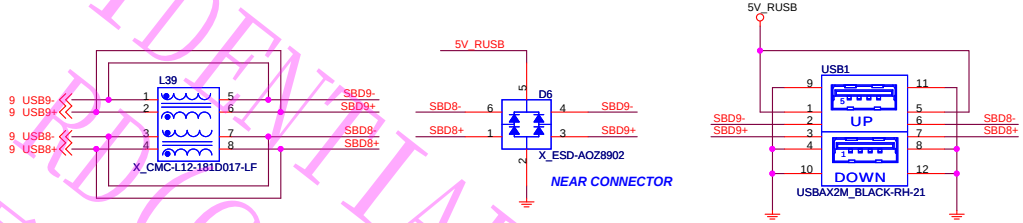
NEAR USB REAR CONNECTOR



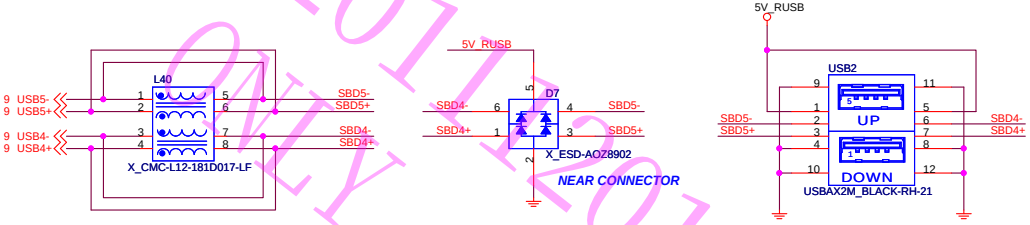
REAR USB PORT 10,11 (With LAN)



REAR USB PORT 8,9



REAL USB PORT 4,5



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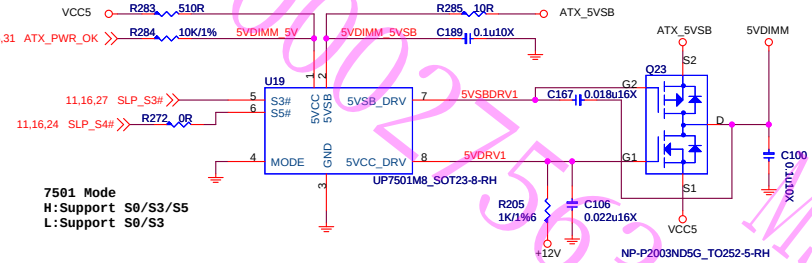
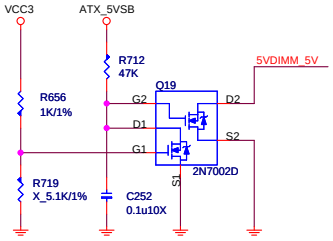
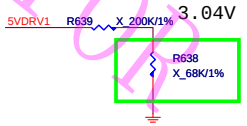
5VDIMM FOR DDR



7501 Mode
H:Support S0/S3/S5
L:Support S0/S3

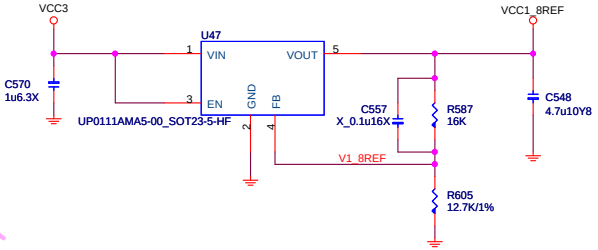


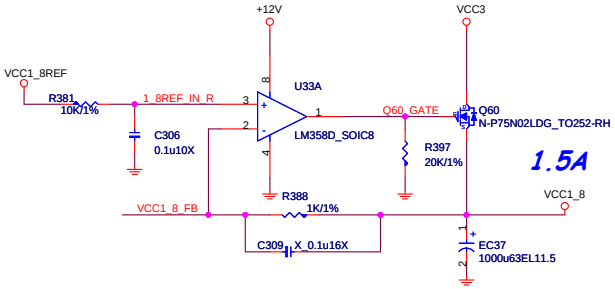
Patch coolermaster 700w power sequence



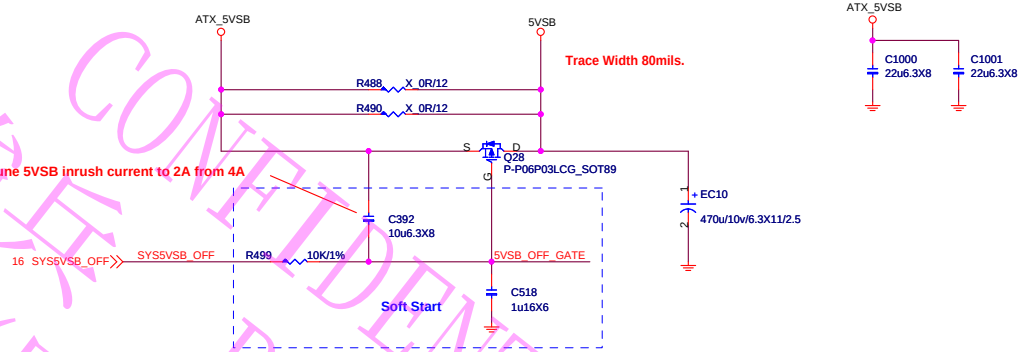
56K-->68K縮少料件

VCC1_8REF



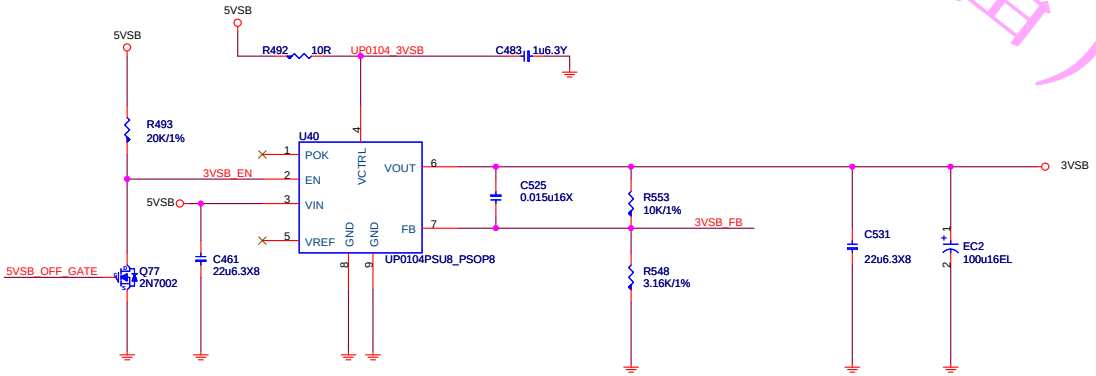


5VSB Power Switch



3VSB

3VSB supply to PCH and other device.
Turn off when Deep S3/S5 by 5VSB off.



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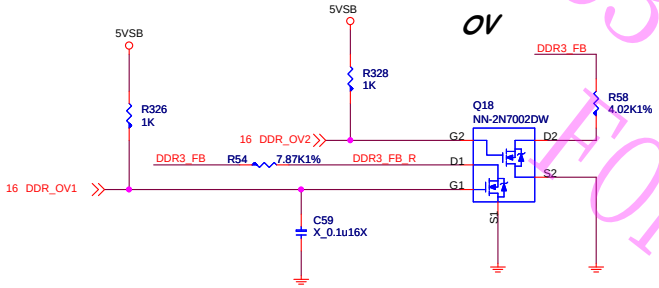
DDR3_1.5V 4.5A+6A+1A+8A=19.5A

4.5A FOR CPU
6A FOR 2DIMM
1A FOR DDR VTT
8A FOR PCH

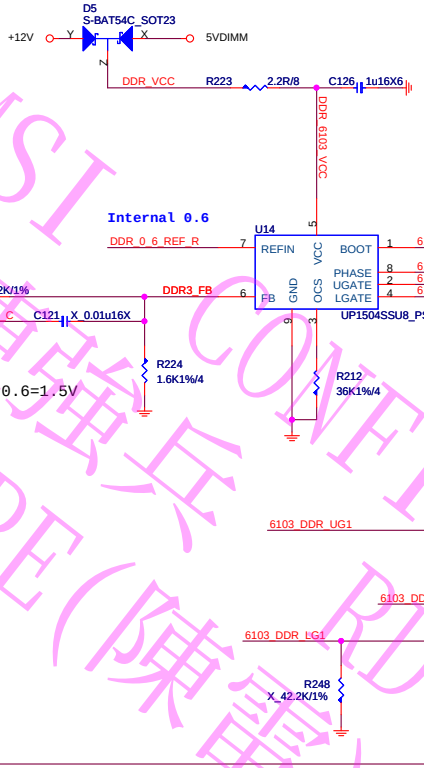


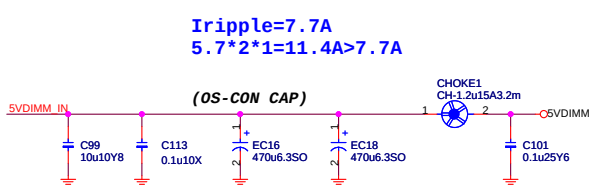
OV

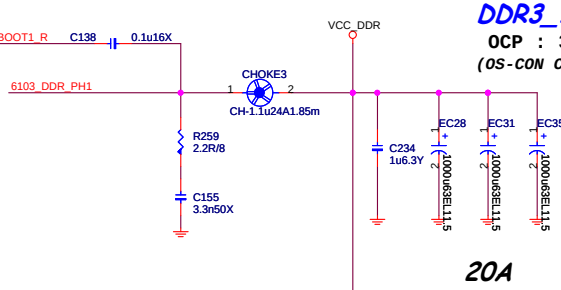
*Default 1.5V

DDR_OV	1.35V	1.5V	1.65V	1.8V
DDR_OV1	Low	High	Low	High
DDR_OV2	Low	Low	High	High

DDR_OV1 = GPIO01(S/IO)
DDR_OV2 = GPIO02(S/IO)

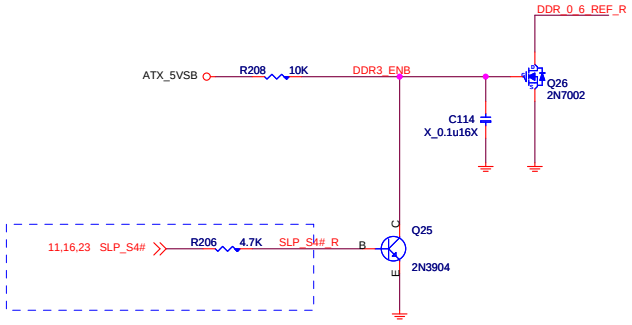






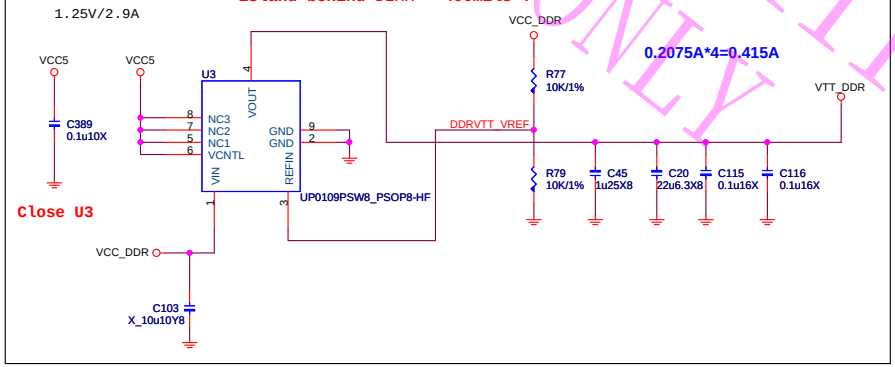
DDR3_1.5V
OCP : 30A
(OS-CON CAP)
20A

P.S. Only for meet Intel power down sequence.



DDR VTT Power

To CPU Copper trace width > 250mils , Fill island behind DIMM > 400mils .



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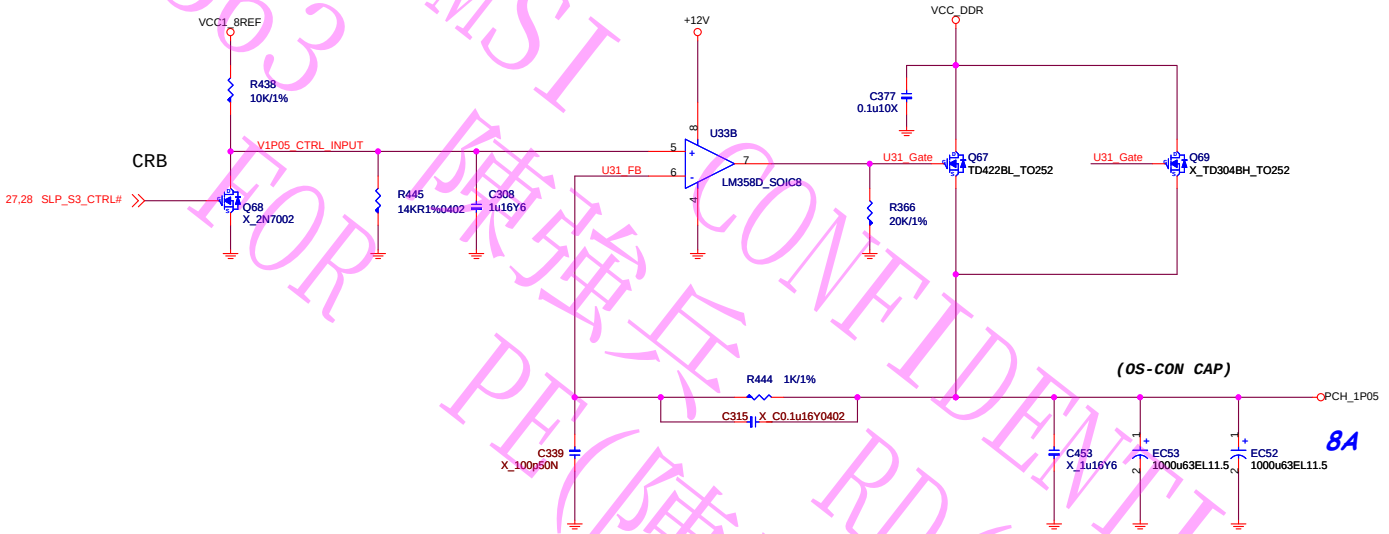
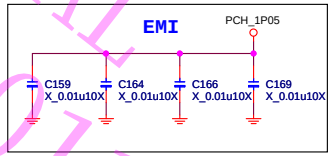
PCH Power:1.05V





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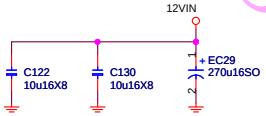
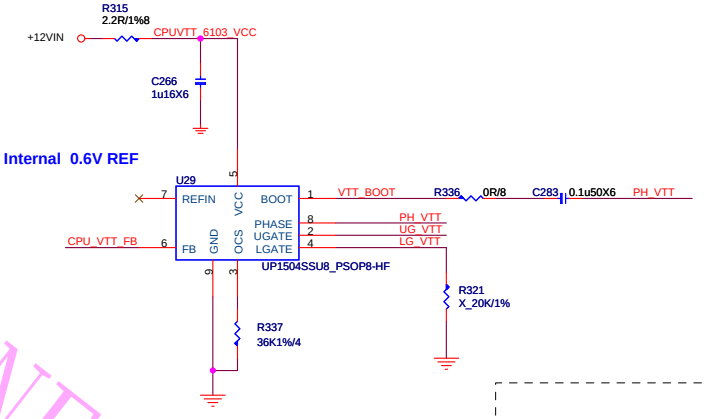
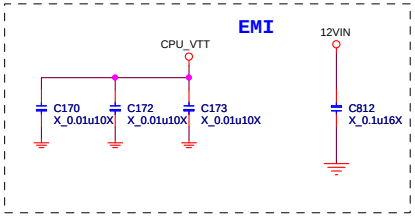
CPU_VTT:1.05/1.00

CPU VTT 8.5A + SA Core =8.8A =17.3A

Iripple=1.92(vtt)+1.88(sa)
5*1=5A>3.8A

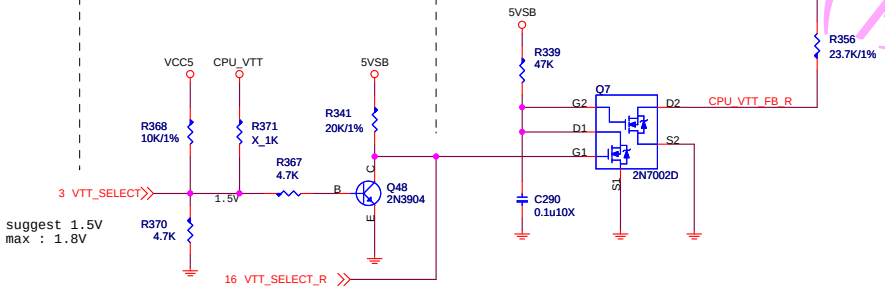


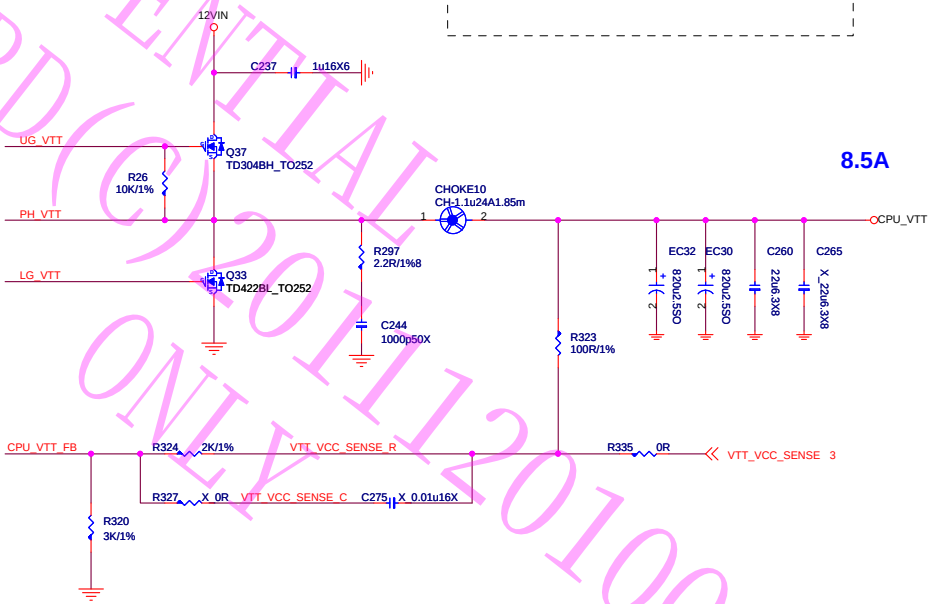




VTT_SELECT	
Low	1.0V
High	1.05V

VTT_SELECT Table	
Low	1.05V
High	1.0V





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CPU_SA:0.925/0.85

SA Core =8.8A

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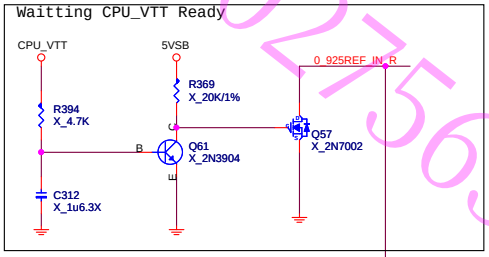

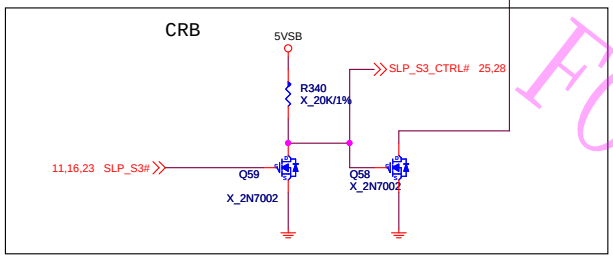

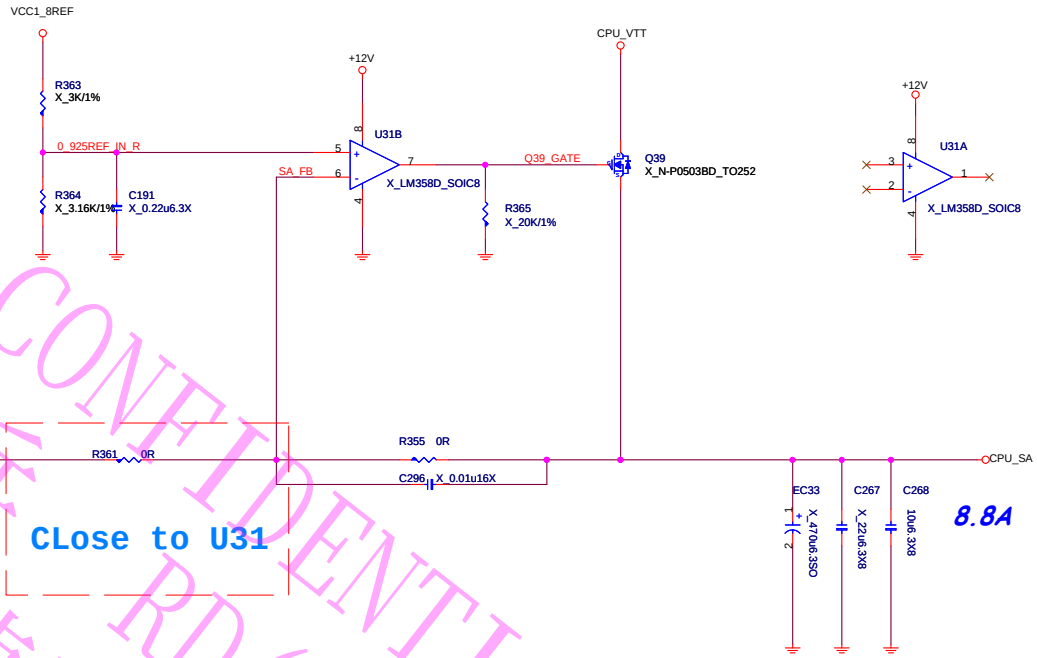

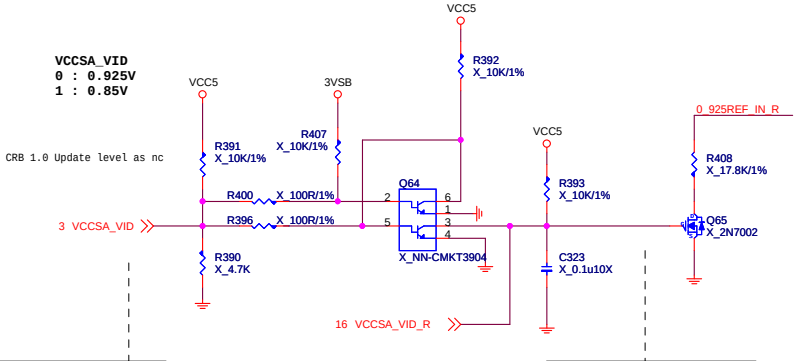

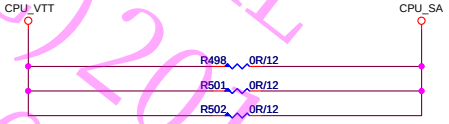

VCCSA_VID	
Low	0.925V
High	0.85V

VCCSA_VID_SIO Table	
Low	0.925V
High	0.85V

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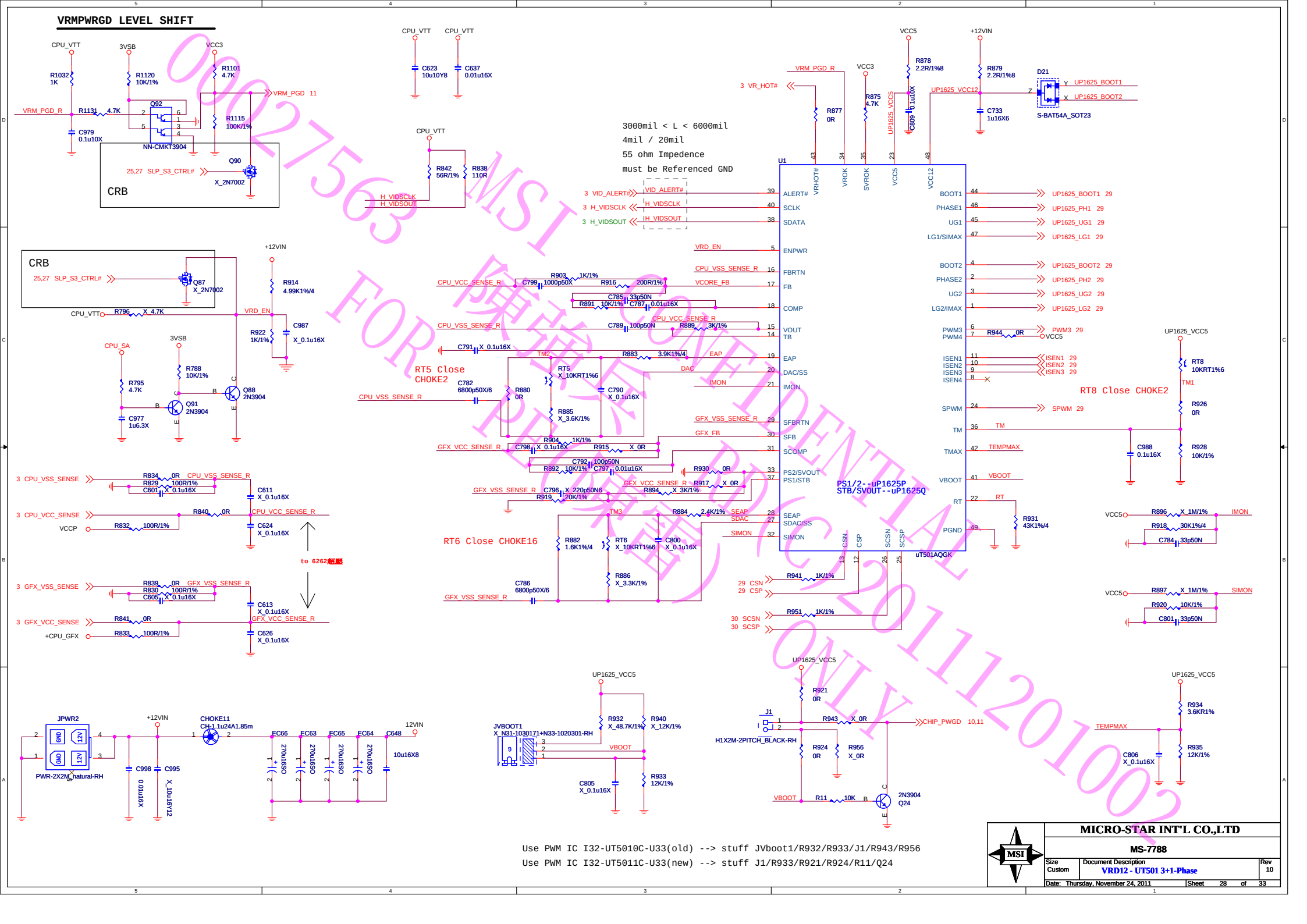
VRMPWRGD LEVEL SHIFT
CRB
RT5 Close CHOKE2
RT8 Close CHOKE2
RT6 Close CHOKE16
3000mil < L < 6000mil
4mil / 20mil
55 ohm Impedence
must be Referenced GND
PS1/2--uP1625P
STB/SV0UT--uP1625Q
uT501AQGK
to 6262相关
Use PWM IC I32-UT5010C-U33(old) --> stuff JVboot1/R932/R933/J1/R943/R956
Use PWM IC I32-UT5011C-U33(new) --> stuff J1/R933/R921/R924/R11/Q24
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MS-7788
VRD12 - UT501 3+1-Phase
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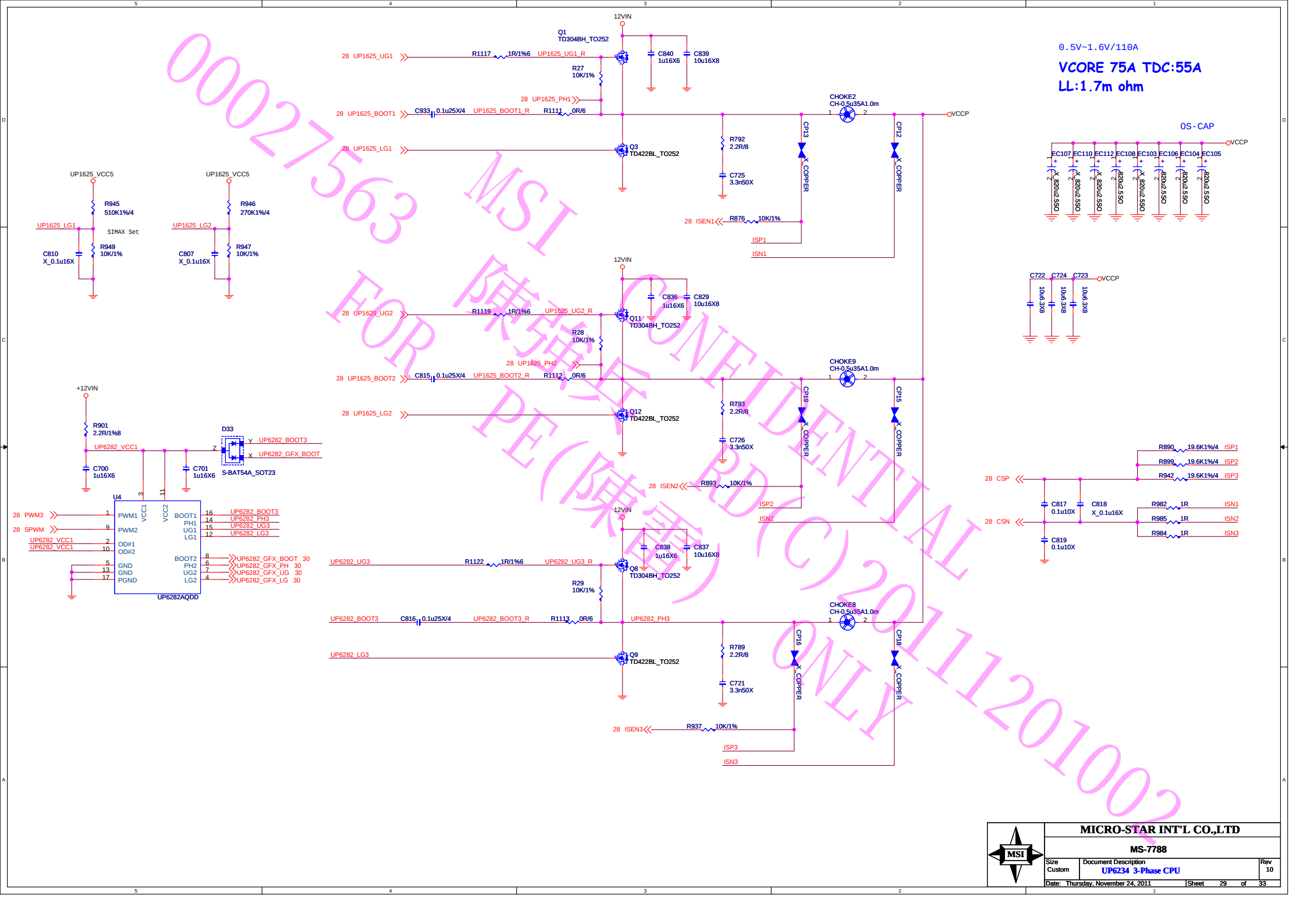
CONFIDENTIAL
FOR PEGATRON ONLY
20111201003



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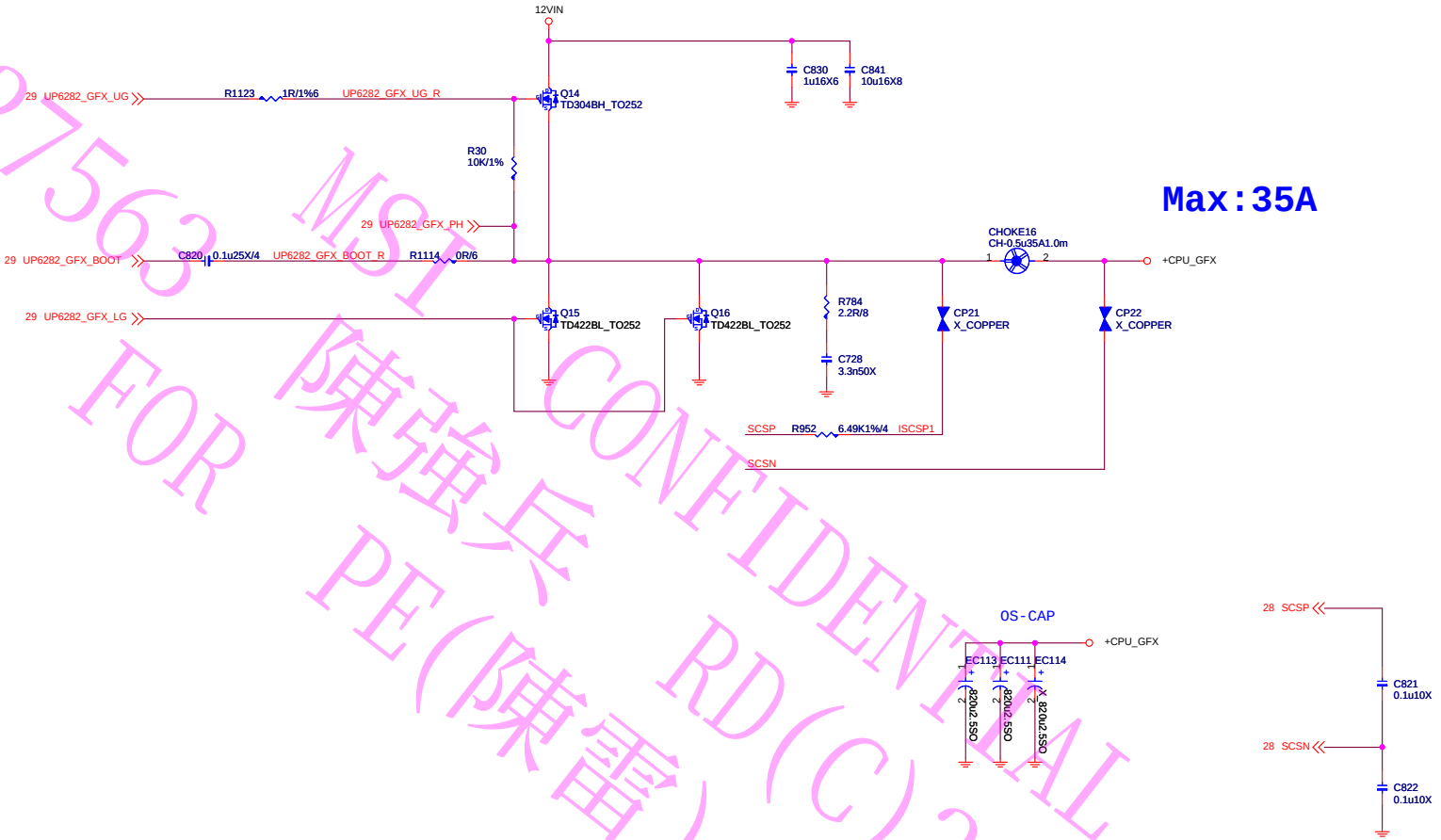

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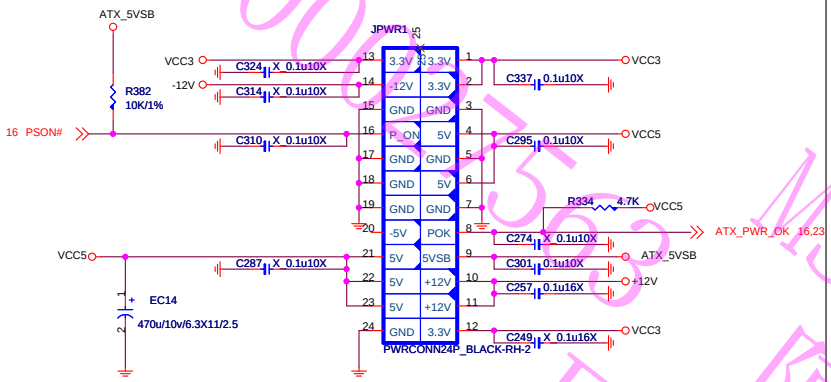
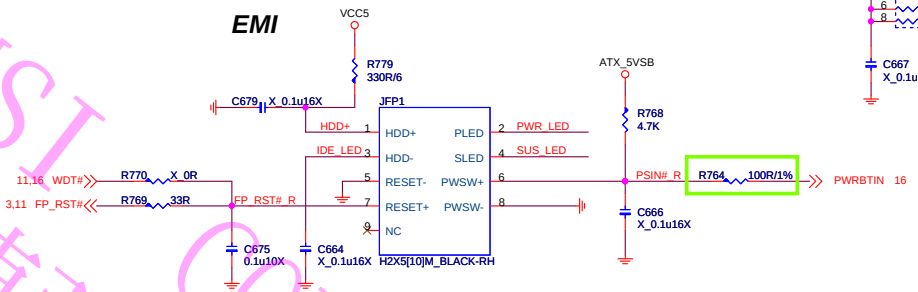
ATX POWER CONNECTOR

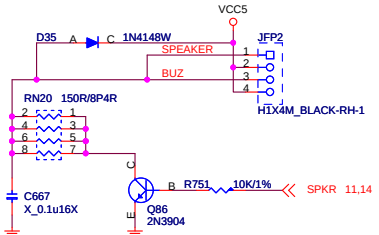


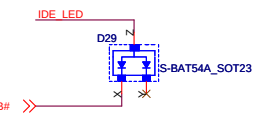
FRONT PANNEL

EMI

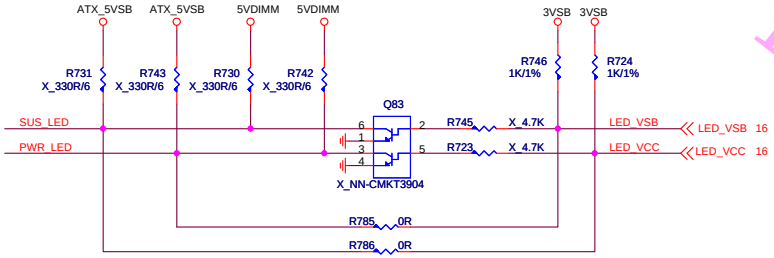


Speaker Pin Header

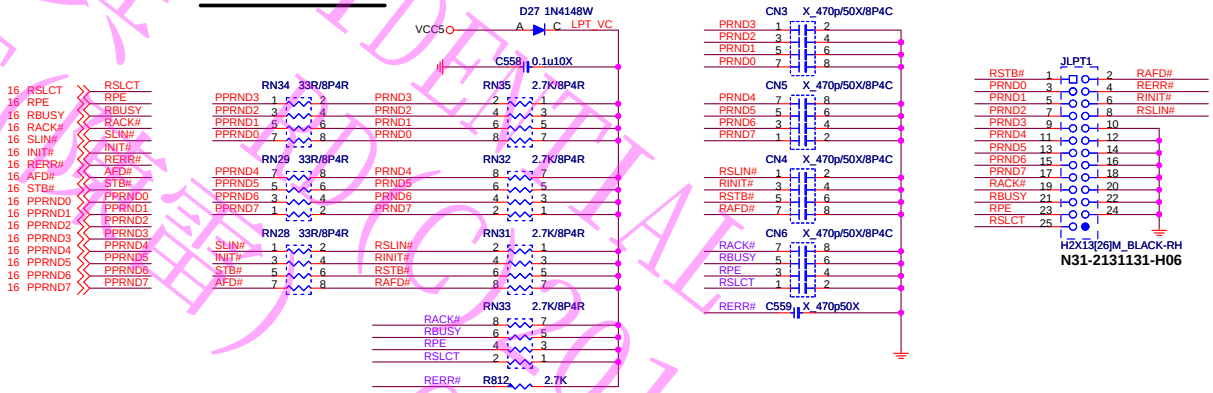


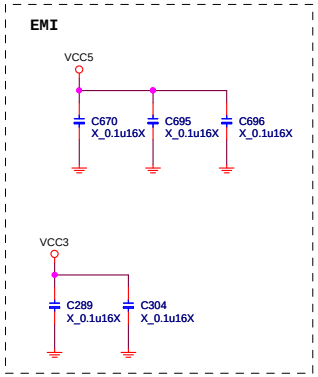


LED (for Fintek 71868)

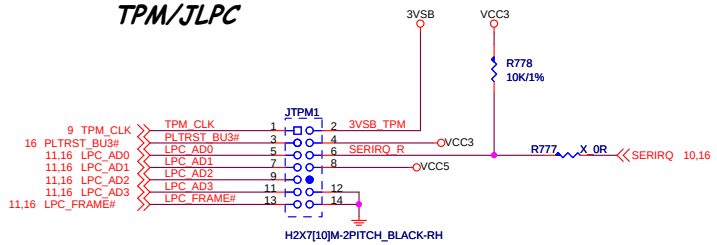


PARALLAL PORT





TPM/JLPC



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MSI	MICRO-STAR INT'L CO.,LTD	
	MS-7788	
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MS-7680-5.1

OPT	Configure	BOM	Function
A	H61M-P30 (B3)	601-7680-300	H61M-P30 (B3),H61 B3,2*DDRIII,1*PCI-Ex16,2*PCI-Ex1,Gb Lan, 4*SATAII,10*USB2.0,HD 8Ch Audio,DVI/D-sub,All Solid Cap,EuP,RoHS
B	H61M-P20 (B3)	601-7680-20S	H61M-P22 (B3),H61 B3,2*DDRIII,1*PCI-Ex16,2*PCI-Ex1,10/100 Lan, 4*SATAII,10*USB2.0,HD 8Ch Audio(3 hole),DVI/D-sub,Half Solid Cap,EuP,RoHS
C	H61M-P25 (B3)	601-7680-310	H61M-P25 (B3),H61 B3,2*DDRIII,1*PCI-Ex16,2*PCI-Ex1,Gb Lan, 4*SATAII,10*USB2.0,HD 8Ch Audio(3 hole),DVI/D-sub,Half Solid Cap,EuP,RoHS
D	H61M-P22 (B3)	601-7680-320	H61M-P22 (B3),H61 B3,2*DDRIII,1*PCI-Ex16,2*PCI-Ex1,10/100 Lan, 4*SATAII,10*USB2.0,HD 8Ch Audio(3 hole),DVI/D-sub,Half Solid Cap,EuP,RoHS

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PCH XDP PWRGD/RESET

HS_PCH1
MEC1
MEC2
HS-0403980-RH

Simulation

SIM2 SIM1 VCC5 X_PIN1*2

Mounting Holes

MH7 MH2 MH4 MH5 MH1 MH8 X_MH001 AGND

Optical Fiducial Marks-120

FM5 FM8 FM2 FM4 FM6 FM7 FM1 FM3 X_FM

LA1 Label 601-7788-A10 LABEL

LA2 AMI_BIOS LABLE BIOS_LABEL

PCB1 7788-10

BAT1_X1 BAT-BCR2032P-RH

RUB1 RUB2 USB X_Rubber

CPU_H1 CPU 鎖座 E21-7557050-L06

PK0-0768051-E36,E&E,23,寶安恩斯邁廠(MSIS),4,Coffee
PK0-0768051-E36,E&E,27,寶安恩斯邁廠(MSIS),4,Coffee
PK0-0768051-G37,精成,23,寶安恩斯邁廠(MSIS),4,Coffee
PK0-0768051-G37,精成,27,寶安恩斯邁廠(MSIS),4,Coffee

MICRO-STAR INT'L CO.,LTD

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MS-7680 5.1 Change to MS-7778 0A

1. 7.1-Channel Audio out change to 5.1-Channel Audio out.
2. Remove PCIe X1 slot (PCI_E3).
3. Remove SPIDF out.
4. Remove Front USB port*2 (JUSB3).
5. JLPT1 2mm pitch change to 2.54mm pitch.

MS-7788 0A Change to MS-7788 1.0

1. modify VT501 COMP & SCOMP not connection to GND.

2. Vcore：E110、EC112、EC103、EC107 >> N.C
   C196、C198、C209、C210、C211 >> N.C
   Vtt：C213、C214、C227、C228、C265 >> N.C
   Vgfx：C109、C203、EC114 >> N.C
   PWM：C782、C786 Chang to 6.8nF；R928 Change to 10K1%
   By Power Solution.

3. Add C634、C642、C794、C795、C802 Empty By Audio codec ALC887 co-lay VT1708S CE.

4. Add C25 Empty By EMI Solution.

5. EC55 & EC56 DIP Footprint Change to SMD DIP Footprint (MSI P/N:C96-1001630-N07) By PM Request.

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